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Stochastic differential equations to model influenza transmission with continuous and discrete-time Markov chains

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ABSTRACT

This study take account into the modeling, mathematical analysis, developing theories and numerical simulation of Influenza virus transmission together with stochastic behavior. We consider the modified five-compartment SEIRT mathematical model to estimate the recent trends and future prediction of basic reproduction number and infection case. Since the basic reproduction number $\mathcal{R}_0$ is not sufficient to predict the outbreak, we apply the discrete-time Markov chain (DTMC), continuous-time Markov chain (CTMC), and Itô stochastic differential equations (SDEs) to calculate the probability of the disease outbreak. Since the H1N1 influenza is a contagious respiratory illness caused by a specific strain of the influenza A virus; which particular strain was designated as H1N1pdm09, with pdm09. We show the positivity and boundedness of solutions and discussed disease-free equilibrium and endemic equilibrium points along with their stability. Based on the results, it reflects that contact patterns affect the dynamics of disease outbreaks. For all the models, the threshold quantity, $\mathcal{R}_0$ is calculated which is the key reason to prove the global and local stability analysis of disease-free equilibrium and endemic equilibrium points. A multitude of numerical outcomes are achieved in order to observe the analytical study. In this study, a more realistic representation of infection dynamics, intervention effects, and transmission behaviors is offered by the convergence of deterministic, CTMC, DTMC, and SDE models. Computational methods are used to validate theoretical results, improving the trustworthiness of the model. The stochastic model allows for an examination of the stochastic character of influenza, which advances our understanding of influenza epidemiology, preventative tactics, and potential treatment results under various circumstances. In the present or future outbreak, such analysis is significant to understand the impact of various parameters and their stochastic behavior in the contagious model to prevent such type disease outbreaks.

1. Introduction

A random process, also known as a stochastic process, is a probability model that depicts the temporal history of a random phenomena; that is, random processes that change randomly over time. A random process, also known as a stochastic process, is a probability model that depicts the temporal history of a random phenomena; that is, random processes that change randomly over time. In a stochastic model, the parameters or state variables are random and their outcomes are not predefined [1–4]. Multidisciplinary research is today considered the most crucial tool for ensuring that all people have normal, healthy lives in our society. Mathematical prediction modeling has shown to be one of the most important techniques for studying and understanding the behavior of epidemics across time. It helps decision-makers make important choices and schedule their days ahead of time. A type of the influenza virus called influenza H1N1, sometimes referred to as swine flu, may infect both people and pigs. Its classification as an influenza subtype The designation H1N1 is applied to a virus that possesses the surface proteins neuraminidase type 1 and hemagglutinin type 1. In 2009, the H1N1 virus sparked a pandemic that garnered much media attention [5,6]. The CDC reports that respiratory droplets are released when an infected person coughs, sneezes, or talks, and these droplets primarily spread the H1N1 virus from one person to another. Nearby people may breathe in these droplets, which would allow the virus to enter their respiratory system. Additionally, it can spread through touching a surface that has the virus on it and then coming into contact with the mouth, nose, or eye. Recently, WHO reported that, the virus might spread inadvertently since it is contagious even before symptoms show. Individuals infected with H1N1 are usually contagious for up to seven days following illness onset and up to one day prior to symptoms appearing. But the length of contagiousness can vary from person to person, particularly in those with compromised immune systems [7–9].

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Throughout human history, pandemics and epidemics have repeatedly decimated people, drastically altering course and often leading to the collapse of whole civilizations. Public health depends on mathematical models that explain the dynamics of infectious diseases because they provide insight into the use of workable and effective disease-control measures. A sickness that resembles influenza and is marked by uncomfortable physical symptoms usually lasts for two to seven days before going away on its own if there are no complications from other illnesses. The immune system of the person has a major impact on how severe the sickness is. The cornerstone for respiratory viral transmission is the generation of virus-containing aerosols and airborne particles that happen during normal speaking and breathing activities. Sneezing is a better way to get rid of viruses from the nasal cavity, especially if it is coupled with increased mucus production from an illness [5–7] .

After being found in both Mexico and the United States, the distinct strain of influenza A (H1N1) was declared a pandemic by the World Health Organization (WHO) on June 11, 2009. People infected with A (H1N1) are not contagious and do not show any symptoms throughout the incubation period. A patient may be infectious from the day before symptoms develop to seven days after they do, and the A (H1N1) virus can incubate for one to four days. Numerous symptoms, including cough, nausea, diarrhea, fever, chills, headache, sore throat, muscular pains, runny nose, shortness of breath, and joint pain, are caused by the influenza A (H1N1) virus [10,11]. Despite the fact that a large number of older people had been exposed to a similar strain of the H1N1 virus in the past and were thus immune to it, the pandemic claimed the lives of over 200,000 people globally [9,12,13].

A vital defense against influenza is the yearly influenza vaccine. The virus's quick transformation, meanwhile, presents a serious obstacle. It is possible that a vaccination that lasts for a year will not offer adequate protection the next year. Moreover, the vaccine's effectiveness may be lowered by the phenomena known as antigenic drift, which can happen after it has been developed. As a result, influenza epidemics are more frequent, especially in high-risk groups [10,11].

It takes more than simply immunization to stop the spread of the flu and other infectious illnesses. Additional preventative measures are also necessary. These include avoiding direct contact with ill people, concealing coughs and sneezes to maintain appropriate respiratory hygiene, and regularly washing your hands to maintain excellent hand hygiene [10,11]. It is possible to decrease the impact of infectious illnesses and the burden they inflict on the population in affected regions by implementing a multifaceted strategy that includes vaccination along with these preventative measures.

Influenza has become a complex parasitic disease in the world. H1N1 can pass from pigs to people in addition to person-to-person transmission. Transmission may occur through direct contact with infected pigs or locations polluted by swine fluids or waste. However, the main cause of H1N1 outbreaks is persistent human-to-human transmission. It is crucial to maintain proper respiratory hygiene if you want to stop the spread of H1N1. This means using an elbow or tissue to cover your mouth and nose when you cough or sneeze, as well as disposing of used tissues properly. Regular hand washing with soap and water or the use of alcohol-based hand sanitizers is crucial to reducing the risk of contamination. Important precautions include staying at home if you experience flu-like symptoms and avoiding close contact with sick people [12–14].

Recent studies showed that, the likelihood of a significant disease epidemic for a certain set of diseases is about $1-\left(\frac{1}{\mathcal{R}_0}\right)^i$, where i is the infectious individuals and $\mathcal{R}_0 > 1$ is the threshold quantity. Stochastic thresholds in the context of several infectious groups are contingent upon the population size and the likelihood of disease extinction within each group. Allen presented the idea of ODE epidemiological dynamics in relation to the topic of stochastic epidemic modeling [15]. The study was especially well-known for stochastic differential equations and continuous-time Markov chains. In order to stop disease epidemics, it is now necessary to identify leading indicators for respiratory diseases, which are a worldwide health burden. It is possible to reduce the number of exposed and diseased people by boosting human immunity. [16–19]. Epidemic model SARS-CoV-2 in Rohingya refugee camp, Bangladesh along with uncertainty and control analysis is discussed in [16]. A modified mathematical modeling on Influenza is developed and stability analysis is focused in recent works [17,18]. Recently Mahmud et al. have studied vaccine efficacy and controlled criterion based on infectious Disease Modeling [19]. In paper [20], Jafari et al. investigates the SIZR model related to Zombie infection outbreaks according to a time-dependent infection rate. In [21] basic mathematical analysis for HIV in deterministic sense is studied by Ahmed et al. Furthermore, Akter et al. have carried out some novel and generic solutions to the space–time fractional modified equal width (MEW) equation using the two variable expansion technique in recent study [22]. Meanwhile, K Dehingia et al. have studied on the dynamics of a financial system with the effect financial information along with several theoretical analysis [23]. A comprehensive study on wave propagation and evolution in a -dimensional spatial–temporal domain is reflected in [24] that is significantly related to realistic behavior of mathematical modeling. Dynamic insights into nonlinear evolution along with effective analytical exploration of a modified width-Burgers equation is studied in [25] by Khater M.M.

Stochastic behavior of the epidemic model is not studied applicability in these works. Most of the studies used small sample sizes, limiting the generalizability of their findings and methodologies employed in traditional ways. The role of CTMC, DTMC, SDE factors was not focused in several recent works. My research aims to fill these gaps by larger samples, modern techniques, and a focus on stochastic behavior of realistic scenarios by adding randomness with deterministic variables.

The behavior of infectious diseases can vary with the passage of time or with the seasons. Because of this, the probability of a disease outbreak may occur periodically and be influenced by seasonal variations and the initial population affected. When the population is big enough, the estimates of the probabilities of a disease breakout from the branching process approximation and CTMC models concur and $\mathcal{R}_0 > 1$. If the number of populations is fewer than 100, the SDE and branching process approximation might not produce a reliable estimate. Using the CTMC model with various parameter values, one may approximate the likelihood of an outbreak for the starting count of COVID-19 infections [26–29].

The classical SIS epidemic model from a deterministic framework with SDE is analyzed recently [26]. To some extent, queuing systems with infinitely many servers and a Markov chain with time-varying transition rate are analyzed in [27]. Deterministic analysis along with stochastic model to describe such random fluctuations and stability based on $\mathcal{R}_0$ is estimated in recent works [28,29].

Another successful method of preventing H1N1 infection is proper treatment. Everyone should get the seasonal influenza vaccine each year, but those who are more vulnerable to complications such as small children, the elderly, pregnant women, and those with underlying medical conditions should have the shot especially. In addition to providing protection against the virus, the seasonal influenza vaccine often provides protection against H1N1. To find the most up-to-date and trustworthy information regarding influenza H1N1, it is recommended that you consult reputable health organizations and governmental sources [30,31]. The Susceptible–Exposed–Infectious–Removal–Treatment (SEIRT) mathematical model, which is used to estimate recent trends, recent probability outbreaks, the extinction of disease, and future predictions of basic reproduction number and infection case, has been modified into five compartments and is deterministic and stochastic.

The objectives of this paper are

- Integrating stochastic models to explore the inherent biological complexity and variability of influenza.
- Providing the foundation of CTMC, DTMC and SDE for influenza propagation.

- Found the probability outbreak concerning the deterministic and seasonal environment.
- Illustrating potential trajectories of influenza transmission, considering both deterministic trends and random fluctuations.

The findings of this paper regarding the goal are

- The study delves into global analysis using Lyapunov mapping, highlighting significant analytical findings.
- The threshold value, ($\mathcal{R}_0$), changes with the kind of modeling, including discrete-time, continuous, and deterministic Markov chains, as well as scenarios including SDEs.
- The interaction among the results of deterministic, CTMC, DTMC, and SDEs reveals a convergence where deterministic outcomes align with stochastic approaches.
- It provides a more accurate depiction of the infection outcomes, interventional effects, and transmission behaviors.
- Exploring theoretical results through validation via computational techniques.
- The stochastic model contributes to a better understanding of influenza epidemiology, prevention strategies, and the possible outcomes of therapies under different conditions by allowing for an exploration of the stochastic nature of influenza.
- According to the study, medication and treatment can be continued for an extended period of time.

Below is the structure of this paper: With regard to the mathematical model, the disease-free equilibrium (DFE) point, basic reproduction number, positivity and boundedness of solutions are all thoroughly explained in Sections 2.1, 3.1 and 3.2, respectively. Furthermore, the impact of the threshold quantity for a disease outbreak is displayed by the calculation of the endemic equilibrium point in Section 3.3. Next, the DFE's local and global stability study is shown in Section 4. In Section 4.3, the results of deterministic formulation is presented numerically. In the Sections Section 5, 6, and 7, we have discussed the formulations of discrete, continuous-time Markov chains (DTMC, CTMC) model, and also the stochastic differential equations (SDEs) model, respectively including Lyapunov stability analysis for SDEs. Additionally, the theoretical results are interpreted and graphically displayed. Lastly, a succinct and useful conclusion is provided in Section 8.

2. Mathematical model for influenza

Chairat introduces mathematical model if influenza transmission in 2017 and keeps us the basis of numerous models for infectious diseases. Chairat has assumed an exponentially dispersed infectious time and treated affected and unaffected people as belonging to a human population with a fixed size [31]. Alcaraz then changed this model, which we took into consideration in this work for our additional research on influenza. Assumed that the Chairat and Alcaraz model's human populations are uniformly mixed [6,30].

The rate of infection transmission from exposed to susceptible is β_1 and healthy human becoming infected by interaction with infected class at a rate β_2. Thus, $\beta_1 ES$ and $\beta_2 IS$ is the infection rate for healthy host from mixing with E and I compartments respectively. Without altering the system's dynamic behavior or generality. Five epidemiological compartments are used to categorize all human populations: susceptible (S), exposed (E), infected (I), recovered (R) and treated (T) individuals. In other words, a modified version of the traditional SEIR epidemic process is utilized to simulate human infection [32]. The SEIRT model describes the flow of individuals between five compartments: susceptible (S), exposed (E), infectious (I), recovered (R), and treated (T). Suppose, the susceptible human get interaction with the exposed and infected compartments. The susceptible population, $S(t)$, increases by the recruitment rate Λ and decreases due to infection with exposed and infectious individuals at rates $\beta_1 E$ and $\beta_2 I$, respectively, as well as natural death at rate μ_S. The exposed population, $E(t)$, grows as susceptible individuals become exposed at the rate $(\beta_1 E + \beta_2 I)S$, and decreases as individuals progress to the infectious state at rate αE or die naturally at rate μ_E. Infectious individuals, $I(t)$, increase as exposed individuals become infectious at rate αE, and decrease due to natural death, disease-induced death, recovery, or isolation at the rates μI, δI, γI, and $\gamma_1 I$, respectively. Recovered individuals, $R(t)$, increase as infectious individuals recover at rate γI and decrease due to natural death at rate μR. Finally, the treated or isolated population, $T(t)$, increases as infectious individuals are moved to treatment or isolation at rate $\gamma_1 I$, and decreases due to natural death at rate μT. This system of equations captures the transitions and interactions between the different compartments over time.

Based on the above, the deterministic ODE system for influenza is given below:

$$\begin{cases} \dfrac{dS}{dt} = \Lambda - (\beta_1 E + \beta_2 I)S - \mu S \\ \dfrac{dE}{dt} = (\beta_1 E + \beta_2 I)S - (\alpha + \mu)E \\ \dfrac{dI}{dt} = \alpha E - (\mu + \delta + \gamma + \gamma_1)I \\ \dfrac{dR}{dt} = \gamma I - \mu R \\ \dfrac{dT}{dt} = \gamma_1 I - \mu T \end{cases} \tag{2.1}$$

along with initial conditions,

$$S(0) = S_0,\ E(0) = E_0,\ I(0) = I_0,\ R(0) = R_0 \ \text{ and } \ T(0) = T_0, \tag{2.2}$$

where the model's total population is determined by

$$N(t) \equiv S(t) + E(t) + I(t) + R(t) + T(t).$$

Here, the total number of humans are $N(t)$, Λ be the recruitment/birth rate of the human population. In this study, we have considered per unit time as weekly basis. Table 1 is presenting all the states variables and all model parameters. We have examined a novel H1N1 serotype influenza A strain in this study. The unusual ability of the influenza virus, an enveloped member of the Orthomyxoviridae family, to vary genetically is founded on two molecular characteristics of the viral family [30,32]. This virus is circulates from infected human to healthy human through interaction, also with cough, sneeze etc. After latent period susceptible population become exposed and exposed compartments progress to the infectious compartments. Then a portion of infected class get recovered and progress to the treatment class for controlling the outbreak of the disease. All the complete is presented in Fig. 1. The influenza compartment diagram is displayed in Fig. 2.

According to the model, the entire human population size satisfies,

$$\begin{aligned} \frac{dN}{dt} &= \Lambda - \mu(S + E + I + R + T) - \delta I \\ &= \lambda - \mu N - \delta I \leq \lambda - \mu N \end{aligned}$$

that asymptotically remains constant, that means,

$$N(t) \longrightarrow \frac{\Lambda}{\mu}$$

Hence, the total population is bounded by $\frac{\Lambda}{\mu}$.

We took into consideration in our study that the recovered population is no longer susceptible for the lowered SEIRT model. Thus, in the event that the patients recover from the illness, their recovery will be permanent. As a result, we have chosen to think about SEIRT type models.

2.1. Positivity and boundedness of solutions

When analyzing an epidemiological model, the positivity and boundedness of the solutions play a crucial role. For $t > 0$ any solution with positive values will remain positive for all the non-negative variables, according to boundedness and positivity. Positive also implies that the population will endure for a long time. Consequently, in order

Table 1
Model state variables and parameters with their descriptions.

Notation	Definition
N	Total number of human population
S	Total number of susceptible population
E	Total number of exposed population
I	Total number of infected population
R	Total number of recovered population
T	Total number of treated population
Λ	Recruitment/birth rate in S class
β_1	Transmission probability of disease from contact (per mixing) with S and E
β_2	Transmission probability of disease from contact (per mixing) with S and I
α	Rate of infection progression from E to I class.
γ	Rate of recovery from I class.
γ_1	Rate of treatment I class to the treated class T.
μ	Natural death(mortality) rate of human (per unit of time)
δ	Disease induced death rate in I compartment (per unit of time)

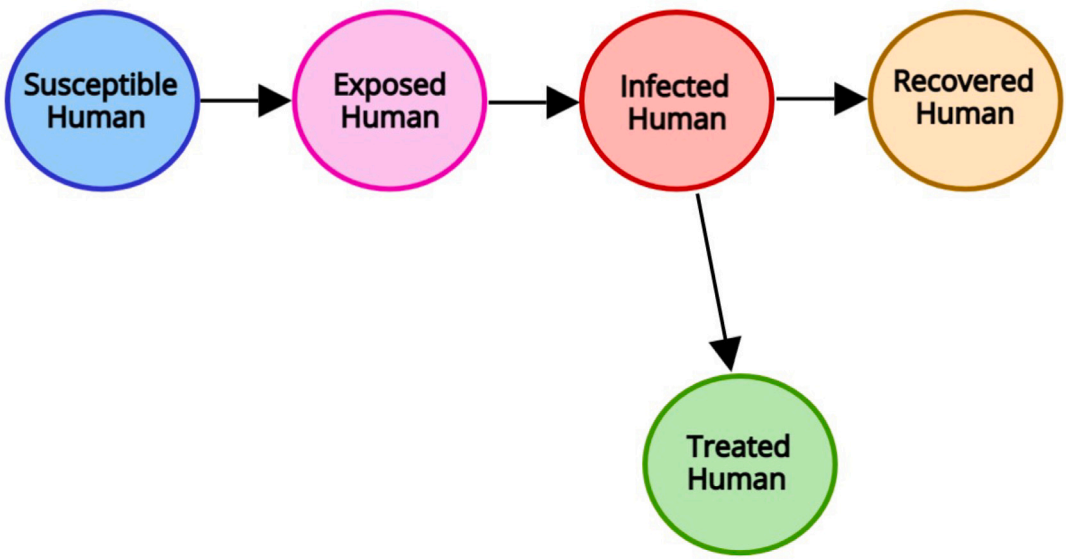


Fig. 1. Transmission cycle in the influenza SEIRT model, when $S(t)$ individuals become exposed to the virus, progress to the infectious stage after a latent period, and then either recover or enter treatment.

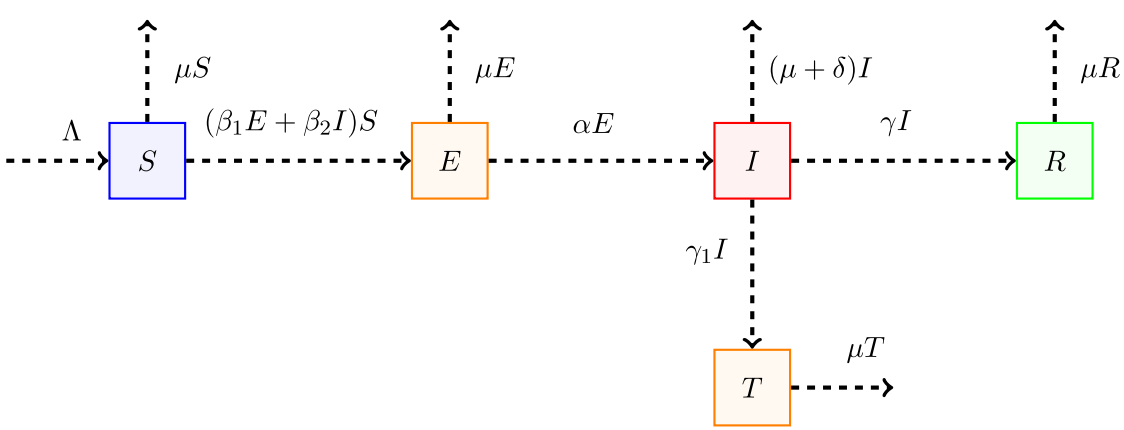


Fig. 2. Compartmental diagram for influenza SEIRT model.

to verify the suggested model's biological validity, we must determine its positivity [8,33–35].

Depending on the compact form's compartment system

$$\mathbf{X}' = F(\mathbf{X}, t), \quad \mathbf{X}(0) = \mathbf{X}_0$$

where, $\mathbf{X} = (X_1, X_2, X_3, X_4, X_5) = (S, E, I, R, T)$. Furthermore, define $k_1 = \frac{\Lambda}{\mu}$. It is comprehensive that F is locally Lipschitz concerning the first argument in $\mathbb{R}^5 \times \mathbb{R}$. Accordingly, using the existence and uniqueness theory for ODEs that are non-linear, it suggests that solutions of $\mathbf{X}(t)$ exist in $(0, T)$ for some T. Here, T might be thought of as the maximum of all these time intervals., and if $T < \infty$, then $\lim_{t \to T^-} \|\mathbf{X}(t)\| \to \infty$. Now, we will convey that, if

$$\mathbf{X}_0 \in \Phi_1 = \{\mathbf{X} \in \mathbb{R}^5 : X_i \geq 0 \text{ for } i = 1, 2, \ldots, 5, \sum_{i=1}^{5} X_i \leq k_1\}$$

Then the set Φ_1 is positively invariant under the direction of flow $\mathbf{X}(t)$ for $t \in (0, T)$ and therefore $T = \infty$, implying that with respect to time the solution exists globally.

Theorem 1 ([35]). *The closed region*

$$\Phi_1 = \{\mathbf{X} \in \mathbb{R}^5 : X_i \geq 0 \text{ for } i = 1, 2, \ldots, 5, \sum_{i=1}^{5} X_i \leq k_1\}$$

is positively invariant beneath the system's generated flow (2.1). Therefore, for initial conditions $\mathbf{X} \in \Phi_1$, solution $\mathbf{X} \in \Phi_1$ globally exists with respect to time.

Proof. We define boundary segments Γ_j^1, $j = 1, 2, 3, 4, 5$.

$$\Gamma_j^1 = \{\mathbf{X} \in \Phi_1 : X_j = 0\} \text{ for } j = 1, 2, 3, 4, 5$$

and $\Gamma_5^1 = \{\mathbf{X} \in \Phi_1 : \sum_{j=1}^{5} X_j \leq k_1\}$

It is evident that, $\partial\Phi_1 = \bigcup_{j=1}^{5} \Gamma_j^1$.

To demonstrate the set's invariance Φ_1, We will demonstrate that for every inward normal (although the inward normal at the junction of one or more Γ_j^1, there is no problem with argument $\mathbf{X}_j' \cdot \mathbf{n}_1 \geq 0$ when $\mathbf{n} = (\mathbf{n}_1)$ is any combination of two or more positive linear inward normals Γ_j^1), $\mathbf{X}_j' \cdot \mathbf{n}_1 \geq 0$ on $\partial\Phi_1$. Evidently, the internal normal on Γ_j^1 for $j = 1, 2, \ldots, 5$ are given by $\mathbf{n}_1^j = \mathbf{e}_1^j = [0, \ldots, 1, \ldots, 0]$, where only non-zero is the j-component, although an inward normal on Γ_5^1 is clearly given by $\mathbf{n}_1^5 = (-1, -1, \ldots, -1)$

It is noted that, on Γ_j^1 for $j = 1, 2, \ldots, 5$:

$$\mathbf{e}_1^1 \cdot \mathbf{X}' = \Lambda - \mu X_1 \geq 0 \text{ for } \mathbf{X} \in \Gamma_1^1$$
$$\mathbf{e}_1^2 \cdot \mathbf{X}' = \beta_1 X_2 X_1 - (\alpha + \mu) X_2 \geq 0 \text{ for } \mathbf{X} \in \Gamma_2^1$$
$$\mathbf{e}_1^3 \cdot \mathbf{X}' = (\mu + \delta + \gamma + \gamma_1) X_3 \geq 0 \text{ for } \mathbf{X} \in \Gamma_3^1$$
$$\mathbf{e}_1^4 \cdot \mathbf{X}' = \mu X_4 \geq 0 \text{ for } \mathbf{X} \in \Gamma_4^1$$
$$\mathbf{e}_1^5 \cdot \mathbf{X}' = \mu X_5 \geq 0 \text{ for } \mathbf{X} \in \Gamma_5^1$$

Considering, $\sum_{j=1}^{5} X_j = k_1$ it implies

$$\frac{dN}{dt} = \Lambda - \mu N(t) - \delta N(t) \leq \Lambda - \mu N(t)$$

Subsequently on Γ_5^1, where $N = k_1$

$$\mathbf{n}_1^5 \cdot X' = -\Lambda + \mu k_1 + \delta I \geq 0$$

for fixed k_1. Therefore, Φ_1 is domain that is positively invariant. Consequently, earlier statements suggest that given beginning conditions $\mathbf{X}_0 \in \Phi_1$, the global existence of the solution $\mathbf{X}(t) \in \Phi_1$ with respect to time is ensured. □

3. Steady states

3.1. Disease-free equilibrium

Equilibrium point is calculated by setting all time derivatives to zero [31,32], we get

$$\begin{cases} \Lambda - (\beta_1 E + \beta_2 I)S - \mu S = 0 \\ (\beta_1 E + \beta_2 I)S - (\alpha + \mu)E = 0 \\ \alpha E - (\mu + \delta + \gamma + \gamma_1)I = 0 \\ \gamma I - \mu R = 0 \\ \gamma_1 I - \mu T = 0 \end{cases} \tag{3.1}$$

Therefore, the equilibrium point free of sickness is

$$P_0 = \left(\frac{\Lambda}{\mu}, 0, 0, 0, 0\right).$$

3.2. Basic reproduction number

In epidemiology, the basic reproduction number, denoted by $\mathcal{R}_0$, is a key parameter that indicates the average number of secondary infections produced by a single infectious individual in a fully susceptible population. If $\mathcal{R}_0 > 1$, the disease can spread in the population, leading to an outbreak. Conversely, if $\mathcal{R}_0 < 1$, the disease will die out over time. The value of $\mathcal{R}_0$ depends on factors such as the transmission rate, the duration of infectiousness, and contact patterns within the population. The next generation matrix (NGM) method is a widely used algorithm to compute the basic reproduction number, $\mathcal{R}_0$. The threshold quantity, $\mathcal{R}_0$, is the dominant eigenvalue of the next generation matrix $\mathcal{K} = \mathcal{F}\mathcal{V}^{-1}$, where $\mathcal{F}$ and $\mathcal{V}$, represents infection matrix and transition matrix respectively. The fundamental reproduction number of the model will be determined at the disease-free equilibrium point (2.1) by using both the Jacobian approach and the next generation approach. Initially, we use the next-generation method for the compartment models in order to determine the fundamental reproduction number. Dieckamn and Heester-beek introduced the next generation matrix in 1990. The spectral radious of this matrix is referred to as the threshold quantity $\mathcal{R}_0$ [32,34].

Regarding the next-generation infectious dynamical model methodology (2.1), suppose

$$\mathcal{F}_i = \begin{pmatrix} (\beta_1 E + \beta_2 I)S \\ 0 \end{pmatrix} \text{ and } \mathcal{V}_j = \begin{pmatrix} (\alpha + \mu)E \\ -\alpha E + (\mu + \delta + \gamma + \gamma_1)I \end{pmatrix} \text{ for } i, j = 1, 2$$

Here, $\mathcal{F}_i$ is the frequency of newly sick people showing up in the ith compartment. $\mathcal{F}_j$ is the remaining terms of transmission, which include deaths, recoveries, and births in the E and I compartment.

At DFE point,we have obtained $S = N = \frac{\Lambda}{\mu}$.

So, the matrix F and V as the Jacobian matrix is constructed of $\mathcal{F}_i$ and $\mathcal{V}_j$, respectively at the equilibrium point free of sickness,

$$F = \begin{pmatrix} \frac{\partial \mathcal{F}_1}{\partial E} & \frac{\partial \mathcal{F}_1}{\partial I} \\ \frac{\partial \mathcal{F}_2}{\partial E} & \frac{\partial \mathcal{F}_2}{\partial I} \end{pmatrix} = \begin{pmatrix} \beta_1 S & \beta_2 S \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \beta_1 S_0 & \beta_2 S_0 \\ 0 & 0 \end{pmatrix}$$

and

$$V = \begin{pmatrix} \frac{\partial \mathcal{V}_1}{\partial E} & \frac{\partial \mathcal{V}_1}{\partial I} \\ \frac{\partial \mathcal{V}_2}{\partial E} & \frac{\partial \mathcal{V}_2}{\partial I} \end{pmatrix} = \begin{pmatrix} \alpha + \mu & 0 \\ -\alpha & \mu + \delta + \gamma + \gamma_1 \end{pmatrix}$$

Therefore,

$$V^{-1} = \begin{pmatrix} \frac{1}{\alpha + \mu} & 0 \\ \frac{\alpha}{(\alpha + \mu)(\mu + \delta + \gamma + \gamma_1)} & \frac{1}{\mu + \delta + \gamma + \gamma_1} \end{pmatrix}$$

Hence, $FV^{-1} = \begin{pmatrix} \frac{S_0 \beta_1}{\alpha + \mu} + \frac{S_0 \alpha \beta_2}{(\alpha + \mu)(\mu + \delta + \gamma + \gamma_1)} & \frac{S_0 \beta_2}{\mu + \delta + \gamma + \gamma_1} \\ 0 & 0 \end{pmatrix}$

The eigenvalues of FV^{-1} are, $\left\{0, \frac{S_0\alpha\beta_2+S_0\beta_1(\mu+\delta+\gamma+\gamma_1)}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)}\right\}$

Thus, the spectral radius of FV^{-1} is,

$$\mathcal{R}_0 = \sigma(FV^{-1}) = \frac{S_0\alpha\beta_2 + S_0\beta_1(\mu+\delta+\gamma+\gamma_1)}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)}$$

It is the fundamental reproduction number ($\mathcal{R}_0$) that we required for the model (2.1).

The reproduction number serves as a threshold condition to examine the stability of the infectious illness model at the point of disease-free equilibrium.

The Model's DFE Point (2.1) is $\left(\frac{\Lambda}{\mu}, 0, 0, 0, 0\right)$. Moreover, $S = \frac{\lambda}{\mu} = N$.

Now, the Jacobian matrix for the model is,

$$J = \begin{pmatrix} a_{11} & -\beta_1 S & -\beta_2 S & 0 & 0 \\ \beta_1 E + \beta_2 I & a_{22} & \beta_2 S & 0 & 0 \\ 0 & \alpha & a_{33} & 0 & 0 \\ 0 & 0 & \gamma & -\mu & 0 \\ 0 & 0 & \gamma_1 & 0 & -\mu \end{pmatrix}$$

where, $a_{11} = -(\beta_1 E+\beta_2 I)-\mu$, $a_{22} = \beta_1 S-(\alpha+\mu)$ and $a_{33} = -(\mu+\delta+\gamma+\gamma_1)$.

Now, computing the Jacobian matrix at the DFE point becomes,

$$J|_{\mathrm{DFE}(P_0)} = \begin{pmatrix} -\mu & -\beta_1 N & -\beta_2 N & 0 & 0 \\ 0 & \beta_1 N - (\alpha+\mu) & \beta_2 N & 0 & 0 \\ 0 & \alpha & -(\mu+\delta+\gamma+\gamma_1) & 0 & 0 \\ 0 & 0 & \gamma & -\mu & 0 \\ 0 & 0 & \gamma_1 & 0 & -\mu \end{pmatrix}. \quad (3.2)$$

Here, $N = \frac{\Lambda}{\mu}$. Consider $|J - \lambda\mathbb{I}| = 0$ ($\mathbb{I}$ is a same-order identity matrix). Find an eigenvalue by expanding the matrix with regard to the fifth column, that is $\lambda_1 = -\mu$. We can identify further eigenvalues by expanding the remaining matrix in terms of the first and fourth columns, that is $\lambda_2 = -\mu$ and $\lambda_3 = -\mu$. The last two eigenvalues can be calculated using the following matrix.

$$J_1 = \begin{pmatrix} \beta_1 N - (\alpha+\mu) & \beta_2 N \\ \alpha & -(\mu+\delta+\gamma+\gamma_1) \end{pmatrix}$$

If the matrix has a positive determinant and a negative trace, then its eigenvalues have a negative real component. We have,

$$\mathrm{Trace}(J_1) = \beta_1 \frac{\Lambda}{\mu} - (\alpha+\mu) - (\mu+\delta+\gamma+\gamma_1) < 0.$$

Thus, the trace of the matrix J_1 is undoubtedly negative. Now,

$$\mathrm{Det}(J_1) = (\alpha+\mu)(\mu+\delta+\gamma+\gamma_1) - \beta_1 S_0(\mu+\delta+\gamma+\gamma_1) - \beta_2 S_0\alpha.$$

According to the requirement that the determinant be positive suggests,

$$\mathrm{Det}(J_1) > 0 \Rightarrow (\alpha+\mu)(\mu+\delta+\gamma+\gamma_1) > \beta_1 S_0(\mu+\delta+\gamma+\gamma_1) + \beta_2 S_0\alpha$$
$$\Rightarrow \frac{\beta_1 S_0(\mu+\delta+\gamma+\gamma_1) + \beta_2 S_0\alpha}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)} < 1.$$

One way to rewrite that condition is as $\mathcal{R}_0 < 1$, Consequently, the fundamental reproduction number is provided by

$$\mathcal{R}_0 = \frac{\beta_1 S_0(\mu+\delta+\gamma+\gamma_1) + \beta_2 S_0\alpha}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)}.$$

In the infectious disease transmission process that we have modified in our model, there involve two stages :

1. Transmission from exposed compartment to susceptible compartment by β_1 rates.
2. Transmission from infected compartment to susceptible compartment by β_2 rates.

For every step of transmission, the reproduction number is the average number of new infection cases; an exposed or infected human can be the initial infection case. Moreover, the fundamental reproduction number $\mathcal{R}_0$ can be presented as a sum of the transmission from exposed to susceptible and infected to susceptible:

$$\mathcal{R}_0 = \frac{\beta_1 S_0(\mu+\delta+\gamma+\gamma_1)}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)} + \frac{\beta_2 S_0\alpha}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)}$$
$$= \frac{\beta_1 S_0}{\alpha+\mu} + \frac{\beta_2 S_0\alpha}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)}.$$

3.3. Endemic equilibrium

To locate system's equilibria (endemic equilibria) (2.1) where at least one of the system's compromised components (2.1) is non-zero, The following steps are required of us. To examine the model's endemic equilibrium state, we obtain,

$$\frac{dS}{dt} = 0, \ \frac{dE}{dt} = 0, \ \frac{dI}{dt} = 0, \ \frac{dR}{dt} = 0, \text{ and } \frac{dT}{dt} = 0.$$

Suppose, $P^* = (S^*, E^*, I^*, R^*, T^*)$ symbolizes any arbitrary endemic equilibrium in the model's framework. (2.1). Thus, system (2.1) can be represented as,

$$\begin{cases} \Lambda - (\beta_1 E^* + \beta_2 I^*)S^* - \mu S^* = 0 \\ (\beta_1 E^* + \beta_2 I^*)S^* - (\alpha+\mu)E^* = 0 \\ \alpha E^* - (\mu+\delta+\gamma+\gamma_1)I^* = 0 \\ \gamma I^* - \mu R^* \\ \gamma_1 I^* - \mu T^* \end{cases} \quad (3.3)$$

From the fourth and fifth equation of system (3.3) we have,

$$R^* = \frac{\gamma I^*}{\mu}, \text{ and } T^* = \frac{\gamma_1 I^*}{\mu}.$$

Again, from the third equation of the system (3.3) we have,

$$E^* = \frac{(\mu+\delta+\gamma+\gamma_1)I^*}{\alpha}.$$

by substituting this in the first equation of (3.3) we have,

$$S^* = \frac{\Lambda\alpha}{\beta_1(\mu+\delta+\gamma+\gamma_1)I^* + \alpha\beta_2 I^* + \alpha\mu}.$$

If $I^* \neq 0$, then after replacing S^* and E^* in the system's second Eq. (2.1) at steady state, we obtain,

$$\frac{(\beta_1(\mu+\delta+\gamma+\gamma_1) + \beta_2\alpha I^*)\Lambda\alpha}{\alpha\beta_1(\mu+\delta+\gamma+\gamma_1)I^* + \alpha^2\beta_2 I^* + \alpha^2\mu} - \frac{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)I^*}{\alpha} = 0$$
$$\Rightarrow \frac{(\beta_1 a_1 I^* + \beta_2\alpha I^*)\Lambda\alpha}{\alpha\beta_1 a_1 I^* + \alpha^2\beta_2 I^* + \alpha^2\mu} - \frac{a_2 a_1 I^*}{\alpha} = 0.$$

Here, we have let $a_1 = (\mu+\delta+\gamma+\gamma_1)$ and $a_2 = (\mu+\alpha)$ which yields,

$$(\alpha\beta_1 a_2 a_1^2 + \alpha^2\beta_2 a_2 a_1)I^{*2} + (\alpha^2\mu a_2 a_1 - \Lambda\alpha^2\beta_1 a_1 + \beta_2\alpha^3\Lambda)I^* = 0.$$

Consequently, following some computations, the quadratic equation:

$$f(I^*) = P_1 I^{*2} + P_2 I^* + P_3, \quad (3.4)$$

where,

$$P_1 = \alpha\beta_1 a_2 a_1^2 + \alpha^2\beta_2 a_2 a_1$$
$$P_2 = \alpha^2\mu a_2 a_1 - \Lambda\alpha^2\beta_1 a_1 + \beta_2\alpha^3\Lambda$$
$$P_3 = 0$$

It depicts clearly that, the coefficient P_1 remain always positive, and P_3 can be positive (negative) according to scenario $\mathcal{R}_0$ can be less than (greater than) unity, respectively. Since, $P_1 > 0$, the presence of equation's positive solutions (3.4) will rely on the indication signs of P_2 and P_3. If $\mathcal{R}_0 > 1$, then there are two roots of Eq. (3.4) of which there is a single positive root, which results in a distinct endemic equilibrium. If $\mathcal{R}_0 = 1$, then we have $P_3 = 0$ and there is a unique non-zero solution of Eq. (3.4), $I^* = -\frac{P_2}{P_1}$, which is positive if and only if $P_2 < 0$. If $P_2 < 0$ there is a positive endemic equilibrium for $\mathcal{R}_0 = 1$. Since equilibrium relies continuously on $\mathcal{R}_0$ it demonstrates the existence of an interval of $\mathcal{R}_0$ on which there two equilibria exist that are positive,

$$I_1^* = \frac{-P_2 - \sqrt{P_2^2 - 4P_1P_3}}{2P_1}, \quad I_1^* = \frac{-P_2 - \sqrt{P_2^2 - 4P_1P_3}}{2P_1}.$$

If $P_3 > 0$ and either $P_2 \geq 0$ or $P_2^2 < 4P_1P_3$, there are no positive solutions of Eq. (3.4) and thus there are no endemic equilibria. Now, from the quadratic equation

$$I^*\left\{(\alpha\beta_1 a_2 a_1^2 + \alpha^2\beta_2 a_2 a_1)I^* + \alpha^2\mu a_2 a_1 - \Lambda\alpha^2\beta_1 a_1 + \beta_2\alpha^3\Lambda\right\} = 0.$$

Here, $I^* = 0$ is the trivial solution which determines the disease-free equilibrium point. The non-trivial solution for I^* is found as:

$$I^* = \frac{\Lambda\alpha^2\beta_1 a_1 - \alpha^2\mu a_2 a_1 - \beta_2\alpha^3\Lambda}{\alpha\beta_1 a_2 a_1^2 + \alpha^2\beta_2 a_2 a_1}.$$

From which we got the explicit expression of I^*. Furthermore, the following conclusions are established for the various ranges of their properties [30,32].

Theorem 2. *Based on the analysis above, the model (2.1) has:*

1. *a distinct endemic equilibrium in Φ_1 if $P_3 < 0$ which is comparable to $\mathcal{R}_0 > 1$,*
2. *a distinct endemic equilibrium in Φ_1 if $P_3 = 0$ or $P_2^2 - 4P_1P_3 = 0$,*
3. *two endemic equilibria in Φ_1 if $P_3 > 0$, $P_2 < 0$ and $P_2^2 - 4P_1P_3 > 0$, and*
4. *no endemic equilibrium states, if not any of above cases.*

Proof. An equivalent proof is available in [31,32] Theorem 3.3 and Theorem 4.2, respectively. □

4. Stability analysis

4.1. The local stability analysis (LAS) of DFE

Theorem 3 ([35]). *The system (2.1) has DFE point which is locally asymptotically stable whenever $\mathcal{R}_0 < 1$ and is unstable if $\mathcal{R}_0 > 1$.*

Proof. The system's DFE point evaluation of the Jacobian matrix (2.1) is

$$J(P_0) = \begin{pmatrix} -\mu & -\beta_1 N & -\beta_2 N & 0 & 0 \\ 0 & \beta_1 N - (\alpha+\mu) & \beta_2 N & 0 & 0 \\ 0 & \alpha & -(\mu+\delta+\gamma+\gamma_1) & 0 & 0 \\ 0 & 0 & \gamma & -\mu & 0 \\ 0 & 0 & \gamma_1 & 0 & -\mu \end{pmatrix} \tag{4.1}$$

$J(P_0)$ possesses the subsequent characteristic equation,

$$|J(P_0) - \lambda\mathbb{I}| = 0$$

Where, $\mathbb{I}$ turns into the identity matrix. As a result, the polynomial in λ, (say) can be found as

$$(\lambda+\mu)(\lambda+\mu)(\lambda+\mu)\left[\left\{\lambda + (\mu+\delta+\gamma+\gamma_1)\right\}\left\{\lambda - \left(\beta_1 N - (\alpha+\mu)\right)\right\} - \beta_2\alpha N\right] = 0$$

Three eigenvalues are given by $\lambda = -\mu < 0$, $\lambda = -\mu < 0$ and $\lambda = -\mu < 0$. The rest eigenvalues are:

$$\lambda^2 + \left\{(\mu+\delta+\gamma+\gamma_1) - \beta_1 S_0 + (\alpha+\mu)\right\}\lambda - \beta_1 S_0(\mu+\delta+\gamma+\gamma_1) + (\alpha+\mu)(\mu+\delta+\gamma+\gamma_1) - \beta_2\alpha S_0 = 0$$

After simplifying the above equation, we get,

$$\lambda^2 + C_1\lambda + C_2 = 0$$

where,

$$\begin{aligned} C_1 &= (\mu+\delta+\gamma+\gamma_1) - \beta_1 S_0 + (\alpha+\mu) \\ C_2 &= (\alpha+\mu)(\mu+\delta+\gamma+\gamma_1) - \beta_2\alpha S_0 - \beta_1 S_0(\mu+\delta+\gamma+\gamma_1) \\ &= (\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)\left\{1 - \frac{\beta_1 S_0(\mu+\delta+\gamma+\gamma_1) + \beta_2\alpha S_0}{(\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)}\right\} \\ &= (\alpha+\mu)(\mu+\delta+\gamma+\gamma_1)(1 - \mathcal{R}_0) \end{aligned}$$

it clearly reveals that, $C_1 > 0$ bur $C_2 > 0$ if and only if $\mathcal{R}_0 < 1$. Hence, the infection free point P_0 is locally asymptotically stable if $\mathcal{R}_0 < 1$ and unstable if $\mathcal{R}_0 > 1$, according to the Routh–Hurwitz criterion. □

4.2. The analysis of DFE point's global stability

The global asymptotic stability (GAS) of P_0 is examined according to the value of $\mathcal{R}_0$ applying the Castillo-Chavez method's requirements [32,34] in this section. To demonstrate the GAS of P_0, the conceivable region $\mathbb{C}_1 = \{(S,E,I,R,T) \in \mathbb{C}_1 : S \leq S^0\}$ is used, where $S^0 = \frac{\Lambda}{\mu}$.

Theorem 4 ([34]). *Regarding the model of infectious disease caused by influenza (2.1), the region*
$\mathbb{C}_1 = \{(S,E,I,R,T) \in \mathbb{C}_1 : S \leq S^0\}$ is a set with positive invariance.

Proof. From the system (2.1), we obtain

$$\begin{aligned} \frac{dS}{dt} &= \Lambda - (\beta_1 E + \beta_2 I)S - \mu S \\ &\leq \Lambda - \mu S \end{aligned}$$

Hence, we get,

$$S(t) \leq S^0 - (S^0 - S(0))e^{-\mu t}.$$

If $S(0) \in \mathbb{C}_1$, then $S(0) \leq S^0$. Therefore, we have

$$S(t) \leq S^0 - (S^0 - S(0))e^{-\mu t} \leq S^0$$

for all $t \geq 0$ whether the initial conditions $S(0) \in \mathbb{C}_1$. Hence the proof. □

Theorem 5 ([34]). *For system (2.1), the DFE P_0 is globally asymptotically stable in the region $\mathbb{C}_1 = \{(S,E,I,R,T) \in \mathbb{C}_1 : S \leq S^0\}$ if $\mathcal{R}_0 < 1$.*

Proof. The system (2.1) is written as

$$\begin{aligned} \frac{dZ}{dt} &= \mathbb{K}(Z, I_1) \\ \frac{dI}{dt} &= \mathbb{L}(Z, I_1) \end{aligned}$$

where, $Z = (S,R,T)^T$ and $I_1 = (E,I)^T$.

Now,

$$\mathbb{K}(Z,I_1) = \begin{pmatrix} \Lambda - (\beta_1 E + \beta_2 I)S - \mu S \\ \gamma I - \mu R \\ \gamma_1 I - \mu T \end{pmatrix}, \quad \mathbb{L}(Z,I_1) = \begin{pmatrix} (\beta_1 E + \beta_2 I)S - (\alpha+\mu)E \\ \alpha E - (\mu+\delta+\gamma+\gamma_1)I \end{pmatrix}$$

Here, $Z \in \mathbb{R}^3$ indicates the quantity of uninfected individuals, and $I_1 \in \mathbb{R}^2$ indicates the total number of populations that are infected, including infectious and latent infections. Moreover,

$$\mathbf{A} = \begin{pmatrix} \beta_1 S - (\alpha+\mu) & \beta_2 S \\ \alpha & -(\mu+\delta+\gamma+\gamma_1) \end{pmatrix},$$

$$\hat{\mathbb{L}}(Z,I_1) = \begin{pmatrix} (\beta_1 E + \beta_2 I)(S^0 - S) - (\alpha+\mu)E \\ \alpha E - (\mu+\delta+\gamma+\gamma_1)I \end{pmatrix}$$

At, $I_1 = 0$, $\mathbb{K}(Z,I_1)$ and $\mathbb{L}(Z,I_1)$ becomes,

$$\mathbb{K}(Z,I_1) = \begin{pmatrix} \Lambda - \mu S \\ -\mu R \\ -\mu T \end{pmatrix} \quad \text{and} \quad \mathbb{L}(Z,I_1) = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

where, $\mathbf{A} = D_I\mathbb{L}(Z^*,0)$, $\mathbb{L}(Z,I_1) = \mathbf{A}I_1 - \hat{\mathbb{L}}(Z,I_1)$ and $Z^* = \left(\frac{\Lambda}{\mu},0,0\right)$ where, Z^* denotes the DFE point for (S,R,T).

Solving the system, $\frac{dZ}{dt} = \mathbb{K}(Z,0)$, we get

$$S(t) = \frac{\Lambda}{\mu} + \left(S(0) - \frac{\Lambda}{\mu}\right)e^{-\mu t},$$

Table 2
Model parameter values with detailed descriptions.

Notation	Definition	Value	Souce
Λ	Recruitment/ Birth rate of human	500×10^{-2}	[30,34,39]
β_1	Illness Rate of transmission from susceptible to exposed	0.0055	[6,36]
β_2	Illness Rate of transmission from infected to susceptible	0.0055	[6,7]
μ	Natural death rate of human	0.02	[8,34]
δ	Disease induced death rate	0.01	[13,37,38]
α	Progression rate of infection from exposed to infected	0.25	[17,18]
γ	Recovery rate from infected class	0.08	[17,18]
γ_1	Treatment rate to infected class	0.05	[8,18]

and hence $\lim_{t\to\infty} S(t) = \frac{\Lambda}{\mu} = S^0$. The solutions of $\frac{dZ}{dt} = \mathbb{K}(Z,0)$ do not rely on the starting conditions $S(0)$. Consequently, the solutions' asymptotic nature $S(t)$ are independent regard to the prerequisites in $\mathbb{C}_1$, assuring the equilibrium point's global asymptotic stability,

$$Z^* = \left(\frac{\Lambda}{\mu}, 0, 0\right)$$

and thus the Castillo-Chavez method's first requirement [34] is satisfied. Further in $\mathbb{C}_1$, $S(t) \leq S^0$ and hence by taking the new infection terms, we have

$$\frac{\beta_1}{N}(S^0 - S) \geq 0 \quad \text{and} \quad \frac{\beta_2}{N}(S^0 - S) \geq 0$$

This suggests that

$$\widehat{\mathbb{L}}(Z, I_1) \geq 0$$

which fulfills the Castollo-Chavez method's second requirement. [15]. As a result, P_0 is GAS if $\mathcal{R}_0 < 1$. □

4.3. Deterministic results and discussion

The suitable parameter values for the intended model of influenza infectious illness (2.1) are included in Table 2. In epidemic models, specific average parameter values are chosen to represent real-world biological processes and capture the dynamics of disease spread accurately. These parameters such as transmission rates β_1, β_2, recovery rates γ, and mortality rates μ, reflect key factors like the average duration of infectiousness, the natural course of the disease. Using realistic parameter values that we used by estimation in this study, ensures the model mirrors actual disease behavior, allowing for predictions and interventions [6,13,36–38]. Parameters are estimated from empirical data or past outbreaks to align the model with observed trends and provide reliable outcomes for public health planning. We used this parameters for analyzing the deterministic curve along with random fluctuations. As different waves vary with nations, so we picked the average values by reading several papers. When it comes to managing influenza epidemics, the rate of disease transmission from exposed to susceptible and from infected to susceptible is crucial.

Figs. 3(a)–(e) reflect the solution of (2.1) with parameters determined by the Table's values 2 and consequently, the fundamental circumstances $S(0) = 990$, $E(0) = 5$, $I(0) = 4$, $R(0) = 1$, $T(0) = 0$, and considering that the total population of humans (the host) are $N(t) = 1000$. Specifically, Figs. 3(a)–(e) have the deterministic answers for host populations, respectively. The Figures show that the number that has been recovered and treated is growing gradually, (Fig. 3(d) and (e)), while the infected and exposed population decreases over time. Furthermore, the susceptible population progress reversely with the progression of exposed and infected individuals. In addition, we are concerned about the way the afflicted people behave. In Fig. 3(c), the afflicted subgroup are represented, and it becomes evident that following a specific time frame (60 weeks), the infected population converges to the zero level and approaches parallelly after that. The same scenario is observed for exposed subpopulation. This indicates that, along with time, by increasing the removal and treatment rate, infected rate decreases notably. Thus, after a long time, the infected and exposed subpopulation moves under control.

Moreover, from Figs. 3(a)–(e), we have observed the effect of contact rates β_1, β_2 on the subpopulation susceptible, exposed, infected class. Also we noticed the effect of recovery rates γ and treatment rates γ_1 in recovered and treated subpopulation. In order to visually represent the sensitive effects of the various model parameters on the suggested model, we show the series of $S(t)$, $E(t)$, $I(t)$, $R(t)$, and $T(t)$ at any time t for different values of parameters taken from Table 2. From Fig. 3(a) It is evident that the population that is susceptible is declining when β_1 and β_2 increases to 40% and 20%. On the other hand, for decreasing rate of these contact rates β_1 and β_2, the susceptible population increases.

From Fig. 3(b),(c) we can observe that, number of infected population increases in notable rate, if the rate of transmission β_1 and β_2 increases. $I(t)$ decreases by the decreasing rate of contact rates. The same scenario is observed for exposed class also. The increasing of contact rates β_1 and β_2 by 20% and 40%, the pick level of exposed and infected classes arises and the opposite scenario for β_1 and β_2 decreasing 20% to 40%. Therefore, the infected and exposed population are highly affected by the increasing rates β_1 and β_2.

Fig. 3(d),(e) depict that, the increasing rate of γ_1 by 20% and 40% increases the recovery rates. Also it is noticed that the treatment rate has positive influence on the treatment compartments. The opposite scenario observed for the decreasing rate of γ and γ_1 by 20% and 40%. The density of the population that has been treated and recovered is therefore greatly impacted by the progression of γ and γ. That indicates that, exposed and infected compartments reduce effectively by increasing rate of γ and γ_1.

In the following sections, we will formulate DTMC and CTMC model by applying Monte Carlo simulation. Considering that Monte Carlo simulation is a computational method for simulating and analyzing intricate processes or systems that involve uncertainty and randomness. It bears the name of the well-known Monte Carlo casino in Monaco, which is well-known for its slot machines.

5. Formulation of Discrete-time Markov Chain (DTMC) model

In the context of an epidemic model, Monte Carlo simulation involves running multiple iterations of the model using randomly generated inputs or parameters. Each iteration represents a possible realization of the epidemic dynamics based on the specified assumptions and random variability [36,40–42].

To formulate a Discrete-time Markov Chain (DTMC) for the Susceptible–Exposed–Infected–Recovered–Treatment model, we can define the states and transition probabilities based on the given system of differential equations:

Transition probability matrix : Let us Denote the Transition probability from state X to Y as $\mathbb{P}(X \to Y)$. The transition probabilities can be derived from the given system of differential equations. Assuming the time step is Δt, we can approximate the derivative terms as finite differences. The transition probabilities are as follows:

$$\mathbb{P}(S \longrightarrow S) = 1 + \Lambda\Delta t - (\beta_1 E + \beta_2 I)\Delta t - \mu\Delta t$$

$$\mathbb{P}(S \longrightarrow E) = (\beta_1 E + \beta_2 I)\Delta t, \quad \mathbb{P}(S \longrightarrow I) = 0$$

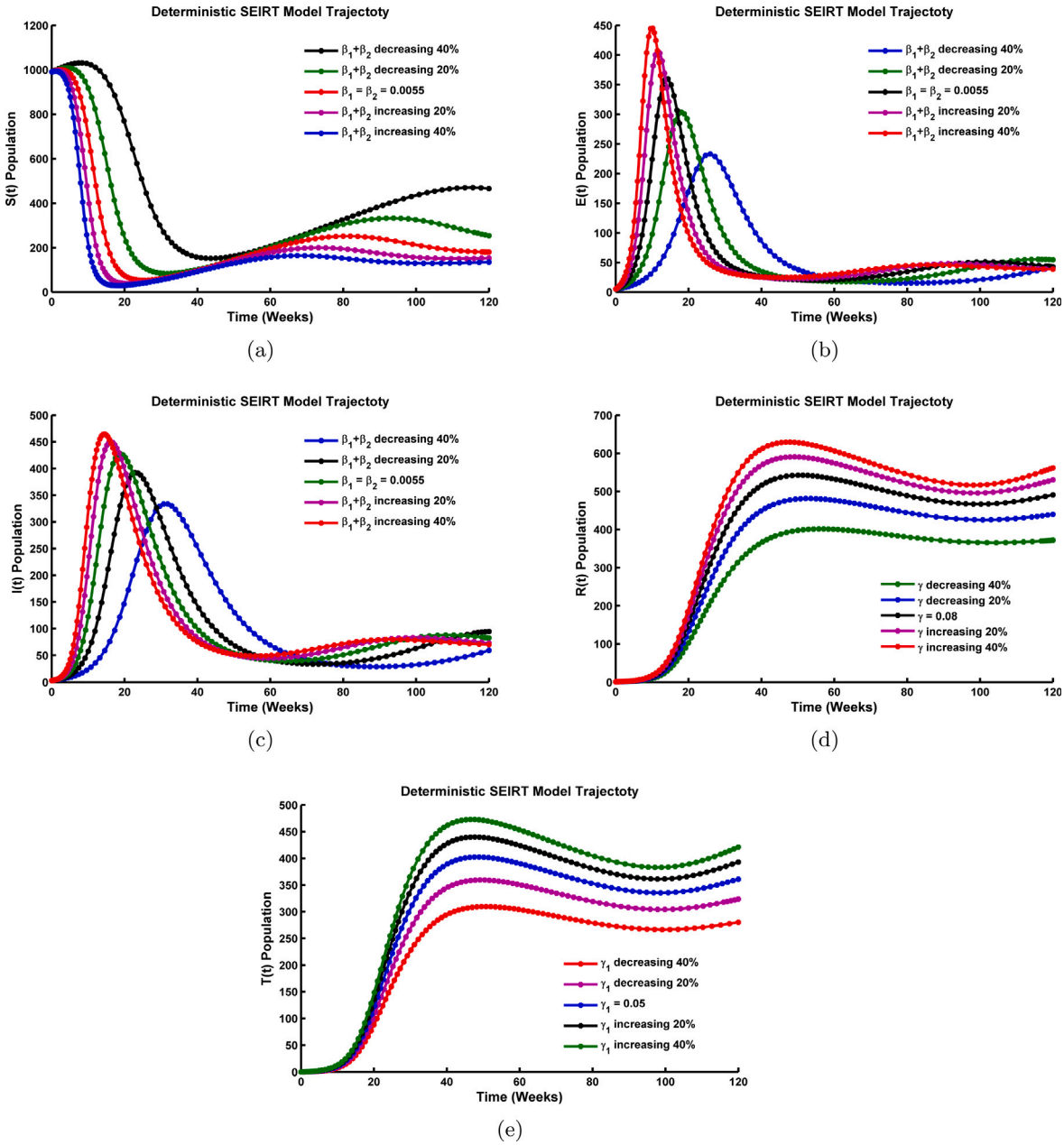


Fig. 3. Dynamics of SEIRT model where (a) Deterministic behavior of susceptible $S(t)$, (b) Deterministic behavior of exposed $E(t)$, (c) Deterministic behavior of infected $I(t)$, (d) Deterministic behavior of recovered $R(t)$, (e) Deterministic behavior of treated population $T(t)$; where, all the mean values of parameters are taken from Table 2.

$$\mathbb{P}(S \longrightarrow R) = 0, \quad \mathbb{P}(S \longrightarrow T) = 0$$
$$\mathbb{P}(E \longrightarrow S) = 0, \quad \mathbb{P}(E \longrightarrow E) = 1 - (\alpha + \mu)\Delta t$$
$$\mathbb{P}(E \longrightarrow I) = \alpha\Delta t, \quad \mathbb{P}(E \longrightarrow R) = 0$$
$$\mathbb{P}(E \longrightarrow T) = 0$$
$$\mathbb{P}(I \longrightarrow S) = 0, \quad \mathbb{P}(I \longrightarrow E) = 0$$
$$\mathbb{P}(I \longrightarrow I) = 1 - (\mu + \delta + \gamma + \gamma_1)\Delta t$$
$$\mathbb{P}(I \longrightarrow R) = \gamma\Delta t, \quad \mathbb{P}(I \longrightarrow T) = \gamma_1\Delta t$$
$$\mathbb{P}(R \longrightarrow S) = 0, \quad \mathbb{P}(R \longrightarrow E) = 0$$
$$\mathbb{P}(R \longrightarrow I) = 0, \quad \mathbb{P}(R \longrightarrow R) = 1 - \mu\Delta t$$
$$\mathbb{P}(R \longrightarrow T) = 0$$
$$\mathbb{P}(T \longrightarrow S) = 0, \quad \mathbb{P}(T \longrightarrow E) = 0$$
$$\mathbb{P}(T \longrightarrow I) = 0, \quad \mathbb{P}(T \longrightarrow R) = 0$$
$$\mathbb{P}(T \longrightarrow T) = 1 - \mu\Delta t$$

Here, the transition probabilities depend on the specific values of the parameters

Λ , β_1, β_2, μ, α, δ, γ, γ_1. The transition probability matrix can be used to model the $S \longrightarrow E \longrightarrow I \longrightarrow R \longrightarrow T$ system as a DTMC, where the state transitions occur based on these probabilities at each time step.

We need to first discritize the continuous-time model. We can use the Euler method to do this, which involves approximating the derivatives in the model with finite differences [38,43]. Let the time step be denoted by Δt, then we have :

$$S(t + \Delta t) \approx S(t) + S'(t)\Delta t$$
$$E(t + \Delta t) \approx E(t) + E'(t)\Delta t$$
$$I(t + \Delta t) \approx I(t) + I'(t)\Delta t$$

$$R(t+\Delta t) \approx R(t) + R'(t)\Delta t$$
$$T(t+\Delta t) \approx T(t) + T'(t)\Delta t$$

Substituting the derivatives from the given model, we get,

$$S(t+\Delta t) \approx S(t) + [\Lambda - (\beta_1 E + \beta_2 I)S - \mu S]\Delta t$$
$$E(t+\Delta t) \approx E(t) + [(\beta_1 E + \beta_2 I)S - (\alpha+\mu)E]\Delta t$$
$$I(t+\Delta t) \approx I(t) + [\alpha E - (\mu+\delta+\gamma+\gamma_1)I]\Delta t$$
$$R(t+\Delta t) \approx R(t) + [\gamma I - \mu R]\Delta t$$
$$T(t+\Delta t) \approx T(t) + [\gamma_1 I - \mu T]\Delta t$$

We can now calculate these equations to construct the transition probability matrix for the DTMC. The states of the DTMC are defined by the values of the variables

$S(t)$, $E(t)$, $I(t)$, $R(t)$, and $T(t)$, at each time step.

Let, $X_t = (S_t, E_t, I_t, R_t, T_t)$ denote the state of the DTMC at time t. The transition probability matrix $\mathbb{P}$ is defined such that $\mathbb{P}_{ij}$ is the probability of transitioning from state i to state j in the one time step. We can compute the transition probabilities using the Euler method as follows:

$$\mathbb{P}_{ij} = \begin{cases} \mathbb{P}\left(X_{t+1} = j | X_t = i\right) = 1 & \text{if } j = i + \Delta x_k \\ \mathbb{P}\left(X_{t+1} = j | X_t = i\right) = 0 & \text{Otherwise} \end{cases}$$

where, Δx_k is a vector of length 5 with a 1 in the kth position and zeros else wise, representing a change of Δx_k in the kth component of the state. Using the equations for the discrete time model, we can write the transition probabilities in terms of the model parameters as follows:

$$\mathbb{P}_{ij} = \begin{cases} 1 + \Lambda\Delta t - (\beta_1 E + \beta_2 I)S\Delta t - \mu S\Delta t & \text{if } j = i + \Delta x_1 \text{ and when } S(t) \geq \Lambda \\ (\beta_1 E + \beta_2 I)S\Delta t & \text{if } j = i + \Delta x_2 \\ 1 - (\alpha+\mu)E\Delta t & \text{if } j = i + \Delta x_2 \\ \alpha E\Delta t & \text{if } j = i + \Delta x_3 \\ 1 - (\mu+\delta+\gamma+\gamma_1)I\Delta t & \text{if } j = i + \Delta x_3 \\ \gamma I\Delta t & \text{if } j = i + \Delta x_4 \\ \gamma_1 I\Delta t & \text{if } j = i + \Delta x_5 \\ 1 - \mu R\Delta t & \text{if } j = i + \Delta x_4 \\ 1 - \mu T\Delta t & \text{if } j = i + \Delta x_5 \\ 0 & \text{otherwise} \end{cases}$$

Here, the transition probabilities are scaled by $\frac{1}{\Lambda\Delta t}$ to ensure that they sum up to 1 for each row of the transition probability matrix. This formulation assumes a fixed time step Δt.

Now, let us rewrite the differential equations in terms of the transition probabilities as follows:

$$\mathbb{P}(S(t+1)|S(t)) = 1 + \Lambda S(t)\Delta t - (\beta_1 E(t) + \beta_2 I(t))S(t)\Delta t - \mu S(t)\Delta t$$
$$\mathbb{P}(E(t+1)|E(t)) = (\beta_1 E(t) + \beta_2 I(t))S(t)\Delta t - (\alpha+\mu)E(t)\Delta t$$
$$\mathbb{P}(I(t+1)|I(t)) = \alpha E(t)\Delta t - (\mu+\delta+\gamma+\gamma_1)\Delta t$$
$$\mathbb{P}(R(t+1)|R(t)) = \gamma I(t)\Delta t - \mu R(t)\Delta t$$
$$\mathbb{P}(T(t+1)|T(t)) = \gamma_1 I(t)\Delta t - \gamma T(t)\Delta t$$

where, Δt represents the time step. The transition probabilities represent the probability of transitioning from one state to another within a single time step. To create a transition probability matrix we can assign these probabilities to the corresponding entries of the matrix based on the states. The matrix will have dimensions (5×5) corresponding to the 5 states. The transition probability matrix ($\mathbb{P}$) can be defined as: (see the equation in Box I).

In the transition probability matrix for the $S \longrightarrow E \longrightarrow I \longrightarrow R \longrightarrow T$ model, each row represents the current state, and each column represents the next state. The rows and columns correspond to the different states in the following order:

1st row represents Susceptible ($S(t)$), 2nd row represents Exposed ($E(t)$), 3rd row represents Infected ($I(t)$), 4th row represents Recovered ($R(t)$), and 5th row represents Under treatment ($T(t)$).

The columns also correspond to the states in the same order. Therefore, the entry in the ith row and jth column represents the probability of transitioning from state i to state j in a single step. For example : let us say, we want to find the transition probability from the Exposed state (E) to the Recovered state (R). This probability would be located in the 2nd Row and 4th column of the transition probability matrix.

So, $\mathbb{P}(E(t+1)|R(t))$ would be the entry in the 2nd Row and 4th column (i.e. $\mathbb{P}_{24}$.)

In summary, each row and column in the transition probability matrix represents a specific state, and the entry at the intersection of row and column represents the probability of transitioning from the state associated with that row in the state associated with that column in a single time step [38,43].

5.1. DTMC results and discussion

For showing the DTMC results graphically, we have taken initial conditions $S(0) = 590$, $E(0) = 5$, $I(0) = 4$, $R(0) = 1$ and $T(0) = 0$. All the mean values of the parameters are taken from Table 2. Fig. 4(a)–(e), representing the compartments of our proposed disease model. From Fig. 4(a), we observed that, the number of susceptible individuals decreases over time as they become exposed or receive treatment. Along weeks 0 to 25 this subpopulation having notable decrease from 600 to 100 (per thousand), after 25 weeks the number of $S(t)$ fluctuates between 100 to 250.

Fig. 4(b) reveals that, number of exposed individuals increases after exposure and then decreases as they either recover or transition to the infected compartment. Along weeks 0 to 18, the exposed population rises peak level around 300, then it gradually decreases. It is observed that along weeks 27 to 120, the exposed population fluctuates between 20 to 150 (per 1000).

We observed from Fig. 4(c) that, the number of infected individuals increases after exposure and then decreases as they either recover or receive treatment. The increasing rate in the exposed population have positive impact on the infected compartments. It is observed that, between 0 to 20 weeks, the number if infected cases arises and strikes peak level around 430. After that, it gradually decreases under control with progression of recovery, and treatment rates. After 60 weeks, the number of infected individuals fluctuates around 20 to 200 (per 1000).

Numerical simulation from Fig. 4(d) describes that, recovered population may have developed immunity through naturally or vaccination. The number of individuals in this compartment typically increases over time as more individuals recover, and it may also serve as an absorbing state, meaning individuals cannot transition out of this compartment. Along weeks 0 to 40 the number of recovered population swiftly increases. After 40 weeks the number of recovered population fluctuates around 250 to 600 (per 1000). That bears a effective impact to reduce the expose and infected density from community.

Fig. 4(e) depicts that, the treatment subpopulation increases if more individuals become infected and require treatment, and decreases as individuals recover. It is noticed that, among 0 to 40 weeks, treatment cases gradually increases. After 60 weeks, the total number of treatment compartments fluctuates around 150 to 350 that reduced the outbreak.

6. Formulation of continuous-time Markov chain (CTMC) model

According to the Markov property, any state variable can be translated to the state i_{n+1} at time t_{n+1} based on the value of the state at that particular moment t_n instead of the process's filtration or background. Inside the CTMC framework. With the Markov property, the process is time-homogeneous, and the discrete-valued state variables assume continuous time [36,41,42].

The deterministic ODE system for influenza will be transformed into the CTMC model.

Let, $\mathfrak{Q}(t) = (S(t), E(t), I(t), R(t), T(t))$.

$$\mathbb{P} = \begin{pmatrix} \mathbb{P}(S(t+1)|S(t)) & \mathbb{P}(S(t+1)|E(t)) & \mathbb{P}(S(t+1)|I(t)) & \mathbb{P}(S(t+1)|R(t)) & \mathbb{P}(S(t+1)|T(t)) \\ \mathbb{P}(E(t+1)|S(t)) & \mathbb{P}(E(t+1)|E(t)) & \mathbb{P}(E(t+1)|I(t)) & \mathbb{P}(E(t+1)|R(t)) & \mathbb{P}(E(t+1)|T(t)) \\ \mathbb{P}(I(t+1)|S(t)) & \mathbb{P}(I(t+1)|E(t)) & \mathbb{P}(I(t+1)|I(t)) & \mathbb{P}(I(t+1)|R(t)) & \mathbb{P}(I(t+1)|T(t)) \\ \mathbb{P}(R(t+1)|S(t)) & \mathbb{P}(R(t+1)|E(t)) & \mathbb{P}(R(t+1)|I(t)) & \mathbb{P}(R(t+1)|R(t)) & \mathbb{P}(R(t+1)|T(t)) \\ \mathbb{P}(T(t+1)|S(t)) & \mathbb{P}(T(t+1)|E(t)) & \mathbb{P}(T(t+1)|I(t)) & \mathbb{P}(T(t+1)|R(t)) & \mathbb{P}(T(t+1)|T(t)) \end{pmatrix}$$

Box I.

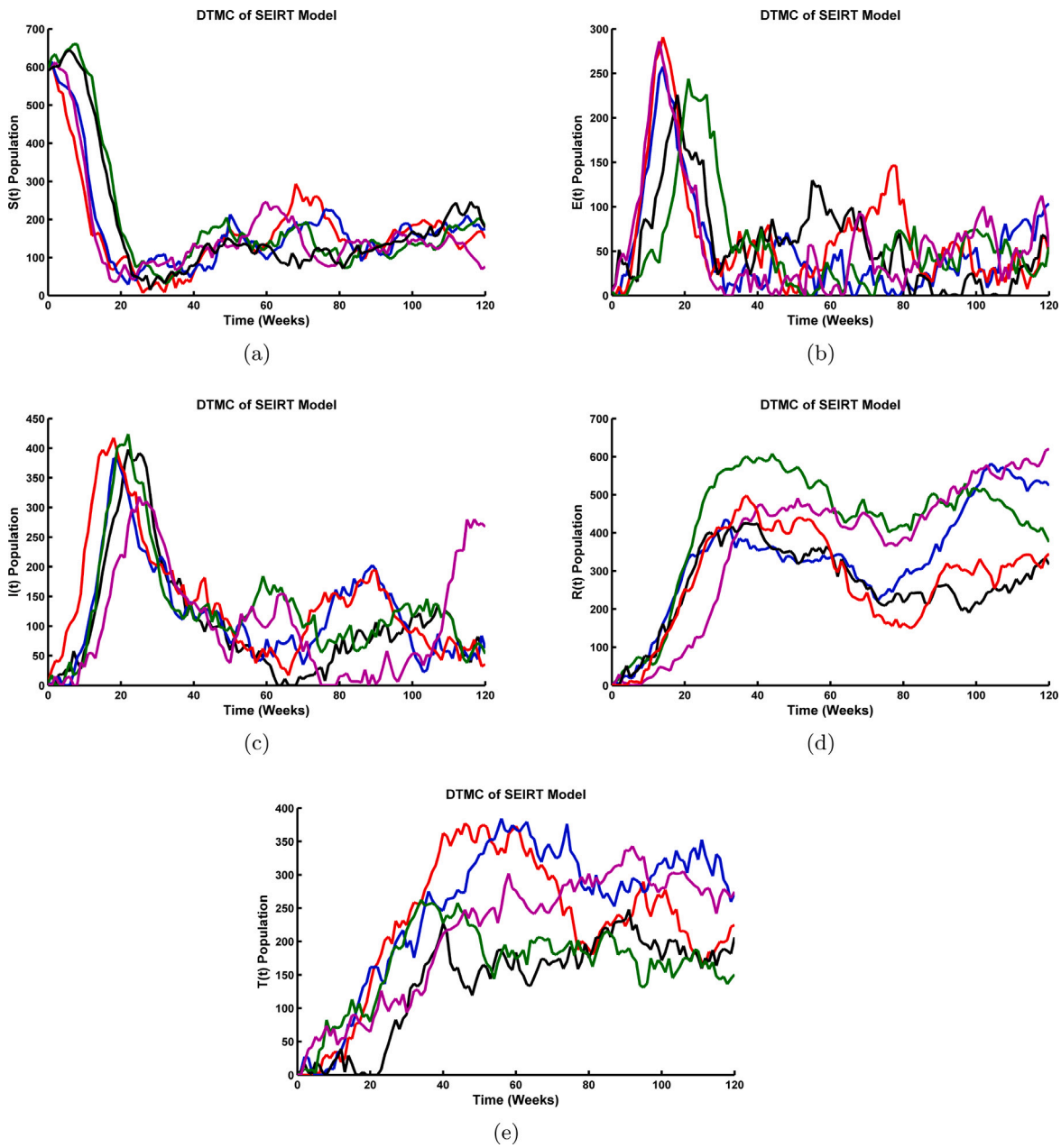


Fig. 4. The curves represent five sample paths in the influenza DTMC models where (a) behavior of susceptible $S(t)$, (b) behavior of exposed $E(t)$, (c) behavior of infected $I(t)$, (d) behavior of recovered $R(t)$, (e) behavior of treated population $T(t)$. The parameter values are $\beta_1 = 0.35,\ \beta_2 = 0.35,\ \gamma = 0.08,\ \gamma_1 = 0.05,\ \alpha = 0.25,\ \mu = 0.02,\ \delta = 0.01$..

In the context of the CTMC model, $S(t),\ E(t),\ I(t),\ R(t),\ T(t)$ acts in a discrete manner across a continuous time period $t \in [0, \infty]$. So,

$$S(t),\ E(t),\ I(t),\ R(t),\ T(t)\ \in\ \{1, 2, 3, 4, \ldots\}$$

Assume that there is a short enough time span $\Delta t > 0$ so that during the time period Δt, at most one incidence occurs. Let,

$$\Delta\mathfrak{Q}(t) = (\Delta S(t), \Delta E(t), \Delta I(t), \Delta R(t), \Delta T(t)),$$

where, $\Delta S(t) = (S(t+\Delta t) - S(t))$, $\Delta I(t) = (I(t+\Delta t) - I(t))$, etc. In other words, it indicates how the system variables have changed during a brief period of time Δt [38,43]. Based on the compartmental system framework (2.1), Table 3 provides a summary of different occurrences and the accompanying transition probabilities for each. The initial occurrence of Table 3 represents the birth of a healthy human, occurrence

Table 3
Potential modifications to the state variables throughout the period Δt that is associated with CTMC model transition probabilities.

Event, i	Description	$\Delta\mathfrak{Q}(\mathbf{t})$	Probabilities $\mathbf{p_i}\Delta(\mathbf{t})+\mathbf{O}(\Delta\mathbf{t})$
1	Healthy Human birth	$(1,0,0,0,0)$	$\Lambda\Delta t+O(\Delta t)$
2	Healthy Human death	$(-1,0,0,0,0)$	$\mu S\Delta t+O(\Delta t)$
3	Infection from exposed to susceptible	$(-1,1,0,0,0)$	$\beta_1 ES\Delta t+O(\Delta t)$
4	Infection from infected to susceptible	$(-1,0,1,0,0)$	$\beta_2 IS\Delta t+O(\Delta t)$
5	Transmission of infection from exposed to infected	$(0,-1,1,0,0)$	$\alpha E\Delta t+O(\Delta t)$
6	Natural death to exposed human	$(0,-1,0,0,0)$	$\mu E\Delta t+O(\Delta t)$
7	Disease induced death to infected human	$(0,0,-1,0,0)$	$\delta I\Delta t+O(\Delta t)$
8	Natural death to infected human	$(0,0,-1,0,0)$	$\mu I\Delta t+O(\Delta t)$
9	Recovery from infected human	$(0,0,-1,1,0)$	$\gamma I\Delta t+O(\Delta t)$
10	Treatment from infected human	$(0,0,-1,0,1)$	$\gamma_1 I\Delta t+O(\Delta t)$
11	Natural death to recovered human	$(0,0,0,-1,0)$	$\mu R\Delta t+O(\Delta t)$
12	Natural death to treated human	$(0,0,0,0,-1)$	$\mu T\Delta t+O(\Delta t)$
13	No change	$(0,0,0,0,0)$	$1-\sum_{i=1}^{12} p_i\Delta t+O(\Delta t)$

5 is the transition of infection from exposed to infected, etc To be more precise, the likelihood of occurrence 3 is equal to,

$$\mathbb{P}(\Delta\mathfrak{Q}(t)) = ((-1,1,0,0,0)|\mathfrak{Q}(t)) = \beta_1 ES\Delta t + O(\Delta t)$$

6.1. CTMC results and discussion

This section aims to examine how the ODE model's (2.1) solution behaves stochastically. For CTMC model, Table 3 displays the transition probabilities for each event that fits the model (2.1). From Figs. 5(a) we can observe the dynamics of susceptible, infected and recovered population, and Fig. 5 reflects the dynamics of exposed and treated population of model (2.1). For each person, they depict the five possible routes. Further, Figs. 6 depict a more detailed perspective of the exposed and infected subpopulation.

Based on the diagram Fig. 5(a), it is expected that the net peak of the infected host will happen between 15 and 30 weeks from now, when there would be a total of 600 and 4 sick people, respectively. The result seen in Fig. 5(a) show that after 23 weeks, the number of affected people gradually declines. At the time span 0 to 23 weeks, infected population swiftly increases, and among 40 to 120 weeks, the random fluctuation of infected individuals varies under 100. Within the time interval 0 to 40 weeks, the recovered individuals gradually increases, and after 40 weeks, the range of random noise in this compartment is 200–380, which is equal to the starting count of recovered individuals 0. On the other hand, the susceptible population remains always in the community. Within the time interval 0 to 20 weeks, it gradually decrease, but when infected population moves under control, the among 40 to 100 weeks, the susceptible population increase lightly and fluctuates among 100 to 120.

Furthermore, within 40 to 60 weeks, the number of infected hosts moves into a meridian position as the host population grows. In addition, from Fig. 5(a), we note that after 25 to 30 weeks, the population's number of infected individuals declines more quickly. Moreover, Fig. 5(b) depicts the scenario of exposed and treated individuals. Within the time span 0 to 17 weeks, the exposed population moves to peak level 250, while initial exposed was 5. After 20 weeks, exposed individuals density started to decrease. Among time interval, 40 to 100, random fluctuation in the exposed class lies within 50 to 70. On the other hand, for time span 0 to 40 weeks, the treated population density gradually increase. After moving to peak level 300 in week 42, the fluctuation in the treatment compartment lies among 150 to 250 up to week 120. That indicates, the increasing rate in treated compartment can control the density of infected and exposed population.

Fig. 6(a) describes the random step in the exposed class by 5 sample paths. The initial conditions in this case are $S(0) = 990$, $E(0) = 4$, $I(0) = 4$, $R(0) = 1$ and $T(0) = 0$. It reflects that, the density rises gradually within weeks 0 to 20 for the high transmission rates β_1 and β_2. After 20 weeks, bu acquiring herd immunity, the swift decreasing scenario occurs in exposed population. From 6(b), the random step infected subpopulation by 5 sample paths reflects. We noticed that, after time span 0 to 17 weeks, infected class strikes to the pick level 800, while the total number of population is 1000. After 40 weeks the random noise bounded among 200 to 300 infected individuals. From Fig. 6, it is noticed that after 40 weeks, by increasing treatment rate, the exposed and infected population density moves under control.

7. Formulation of a SDE model

Next, we will use the CTMC model to obtain the Itô stochastic differential equations (SDEs). Stochastic differential equation (SDE) models are mathematical frameworks used in epidemiology to explain the dynamics and transmission of infectious illnesses. Unlike deterministic models, which assume that the variables evolve continuously and deterministically, SDE models account for randomness and fluctuations in the system [1,2,26,36]. The general form of an SDE model used in epidemiology is:

$$dX(t) = \mu(X(t),t)dt + \sigma(X(t),t)dW(t)$$

where,

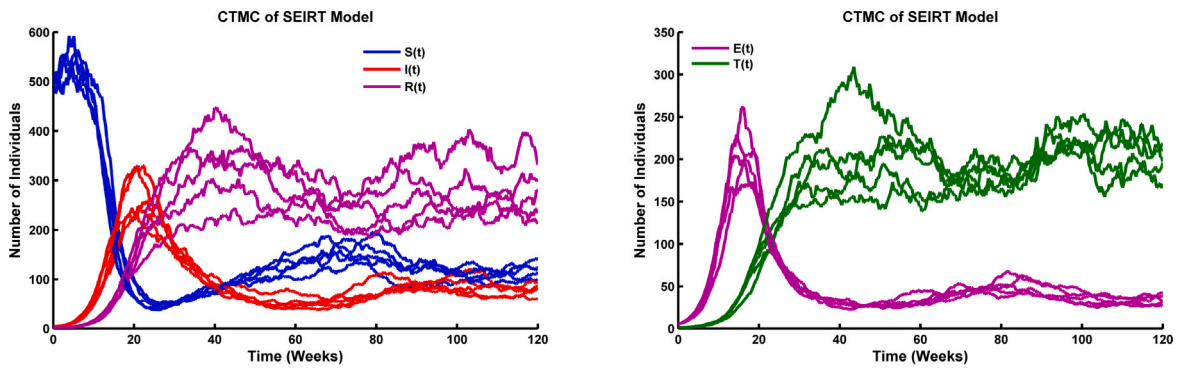
1. $X(t)$ depicts the system's state variables at time t (e.g., the number of infected individuals),
2. $\mu(X(t),t)$ is the deterministic component of the model, describing the average rate of change of $X(t)$ with respect to time,
3. $\sigma(X(t),t)$ is the stochastic component of the model, representing the random fluctuations in the system,
4. $dW(t)$ is a Wiener process or Brownian motion, which represents the random noise in the model.

The parameter and variable notations used in the deterministic system and CTMC model will be applied here. Time and random variables in the SDE model are both continuous and fall inside the set $[0,\infty)$ [15,34]. The system of SDEs leads as:

$$d\mathfrak{Q} = u(t) + \mathcal{G}(t)dW(t)$$

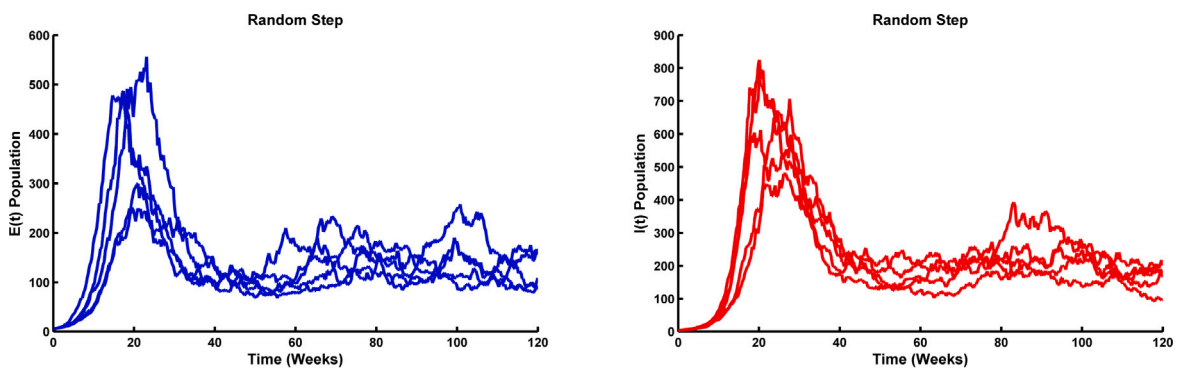
where, $u(t)$ represents the drift vectors, $\mathcal{G}(t)$ is the diffusion matrix which satisfies $\mathcal{G}(t)\mathcal{G}^T(t) = C(t)$, and $W(t)$ is such vector consists of ten separate Wiener or Brownian processes. More precisely, let $\mathcal{G} = (\check{\mathcal{A}}|\check{\mathcal{B}})$ where, $\check{\mathcal{A}}$ and $\check{\mathcal{B}}$ are both 5×5 matrix, correspondingly, as follows:

$$\check{\mathcal{A}} = \begin{pmatrix} \sqrt{\mu N} & -\sqrt{\beta_1 ES} & -\sqrt{\beta_2 IS} & -\sqrt{\mu S} & 0 \\ 0 & \sqrt{\beta_1 ES} & \sqrt{\beta_2 IS} & 0 & -\sqrt{\alpha E} \\ 0 & 0 & 0 & 0 & \sqrt{\alpha E} \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$$



(a) (b)

Fig. 5. Five sample routes are represented by the curves in the influenza CTMC models, where (a) behavior of susceptible, infected and recovered, (b) behavior of exposed and treated population. The parameter values are $\beta_1 = 0.35,\ \beta_2 = 0.35,\ \gamma = 0.08,\ \gamma_1 = 0.05,\ \alpha = 0.25,\ \mu = 0.02,\ \delta = 0.01$..



(a) (b)

Fig. 6. The curves represent five sample paths of random step function where (a) closer view of exposed, (b) closer view of infected population. The parameter values are $\beta_1 = 0.35,\ \beta_2 = 0.35,\ \gamma = 0.08,\ \gamma_1 = 0.05,\ \alpha = 0.25,\ \mu = 0.02,\ \delta = 0.01,$ noise $= 0.08$..

$$\breve{B} = \begin{pmatrix} 0 & 0 & 0 & 0 & 0 \\ -\sqrt{\mu E} & 0 & 0 & 0 & 0 \\ 0 & -\sqrt{(\mu+\delta)I} & -\sqrt{\gamma I} & -\sqrt{\gamma_1 I} & 0 \\ 0 & 0 & \sqrt{\gamma I} & 0 & -\sqrt{\mu R} \\ 0 & 0 & 0 & \sqrt{\gamma_1 I} & -\sqrt{\mu T} \end{pmatrix}$$

Thus, consequently, the influenza infectious illness model's system of SDEs is depicted as:

$$\begin{aligned} dS(t) &= u_1 dt + \sqrt{\mu N} W_1 - \sqrt{\beta_1 ES} W_2 - \sqrt{\beta_2 IS} W_3 - \sqrt{\mu S} W_4 \\ dE(t) &= u_2 dt + \sqrt{\beta_1 ES} W_2 + \sqrt{\beta_2 IS} W_3 - \sqrt{\alpha E} W_5 - \sqrt{\mu E} W_6 \\ dI(t) &= u_3 dt + \sqrt{\alpha E} W_5 - \sqrt{(\mu+\delta) I} W_7 - \sqrt{\gamma I} W_8 - \sqrt{\gamma_1 I} W_9 \\ dR(t) &= u_4 dt + \sqrt{\gamma I} W_8 - \sqrt{\mu R} W_{10} \\ dT(t) &= u_5 dt + \sqrt{\gamma_1 I} W_9 - \sqrt{\mu T} W_{10} \end{aligned} \tag{7.1}$$

where, the column vector $u = (u_1, u_2, u_3, u_4, u_5)^T$ originates from the deterministic ode model directly, which implies,

$$u = \begin{pmatrix} \lambda - (\beta_1 E + \beta_2 I)S - \mu S \\ (\beta_1 E + \beta_2 I)S - (\alpha + \mu)E \\ \alpha E - (\mu + \delta + \gamma + \gamma_1)I \\ \gamma I - \mu R \\ \gamma_1 I - \mu T \end{pmatrix}$$

Moreover, each of the column vector of $\breve{A}$ corresponds to $W_1,\ W_2,\ W_3,\ W_4,\ W_5$ respectively and each of column vector $\breve{B}$ corresponds to $W_6,\ W_7,\ W_8,\ W_9,\ W_{10}$ respectively.

The set of stochastic differential equations that follows can also be used to represent the SEIRT model:

$$\begin{aligned} dS &= [\Lambda - (\beta_1 E + \beta_2 I)S - \mu S]dt + \sigma S dW_1 \\ dE &= [(\beta_1 E + \beta_2 I)S - (\alpha + \mu)E]dt + \rho S dW_2 \\ dI &= [\alpha E - (\mu + \delta + \gamma + \gamma_1)I]dt + \theta S dW_3 \\ dR &= [\gamma I - \mu R]dt + \phi S dW_4 \\ dT &= [\gamma_1 I - \mu T]dt + \psi S dW_5 \end{aligned}$$

where the previous tables list all of the state variables and parameters.

Furthermore, $W_i(t),\ \ i = 1, 2, \ldots, 5$ stand for conventional Brownian motions as independent Wiener processes. The parameters $\sigma,\ \rho,\ \theta,\ \phi,$ and ψ denote the intensities of noise terms that drive variability in each compartment over time. The Lyapunov stability theorem [26,36], for this SDE states as follows:

Theorem 6 ([27,36]). *The stochastic system is stable if there exists a scalar function $V(x(t))$ which satisfies the following conditions:*

(1) $V(x)$ is positive definite, i.e., $V(x) > 0$ for all $x \neq 0$ and $V(0) = 0$.
(2) The time derivative of V along the trajectories of the system is negative semi-definite, i.e., $\frac{dV}{dt} \leq 0$.

Proof. In order to analyze the stability of the stochastic SEIRT model, we will formulate and prove the Lyapunov stability. Let us define the Lyapunov function for the system as follows:

$$V(S,E,I,R,T) = a_1S^2 + a_2E^2 + a_3I^2 + a_4R^2 + a_5T^2$$

where, a_1, a_2, a_3, a_4, a_5 are positive constants and also $S(t)$, $E(t)$, $I(t)$, $R(t)$, $T(t)$ are non-negative. Thus $V(x)$ is certainly positive definite. Hence, first part is proved, Now, we will proceed to the second part.

The time derivative of V can be calculated using Ito's Lemma:

$$\begin{aligned} dV &= \frac{\partial V}{\partial S}dS + \frac{\partial V}{\partial E}dE + \frac{\partial V}{\partial I}dI + \frac{\partial V}{\partial R}dR + \frac{\partial V}{\partial T}dT \\ &= \left(2a_1S[\Lambda - (\beta_1E + \beta_2I)S - \mu S] + 2a_2E[(\beta_1E + \beta_2I)S - (\alpha+\mu)E]\right)dt \\ &\quad + \left(2a_3I[\alpha E - (\mu+\delta+\gamma+\gamma_1)I] + 2a_4R[\gamma I - \mu R] + 2a_5T[\gamma_1 I - \mu T]\right)dt \\ &\quad + \left(a_1S\sigma S + a_2\rho SE + a_3\theta SI + a_4\phi SR + a_5\psi ST\right)dW \end{aligned}$$

where dW represents the stochastic differential terms ($dW_1, dW_2, dW_3, dW_4, dW_5$). Now, we can simplify the above expression as follows:

$$\begin{aligned} dV &= \left(2a_1S[\Lambda - (\beta_1E + \beta_2I)S - \mu S]\right)dt \\ &\quad + \left(2a_2E[(\beta_1E + \beta_2I)S - (\alpha+\mu)E]\right)dt \\ &\quad + \left(2a_3I[\alpha E - (\mu+\delta+\gamma+\gamma_1)I]\right)dt \\ &\quad + \left(2a_4R[\gamma I - \mu R]\right)dt + \left(2a_5T[\gamma_1 I - \mu T]\right)dt \\ &\quad + \left(a_1S\sigma S + a_2\rho SE + a_3\theta SI + a_4\phi SR + a_5\psi ST\right)dW. \end{aligned}$$

Now, let us simplify the expression further by grouping similar terms:

$$\begin{aligned} dV &= \left[2a_1S(\Lambda - (\beta_1E + \beta_2I)S - \mu S) + 2a_2E((\beta_1E + \beta_2I)S - (\alpha+\mu)E)\right. \\ &\quad \left. + 2a_3I(\alpha E - (\mu+\delta+\gamma+\gamma_1)I) + 2a_4R(\gamma I - \mu R) + 2a_5T(\gamma_1 I - \mu T)\right]dt \\ &\quad + \left[a_1\sigma S^2 + a_2\rho SE + a_3\theta SI + a_4\phi SR + a_5\psi ST\right]dW. \end{aligned}$$

Now, let us focus on the drift terms within the square brackets. We can see that each term can be written as the derivative of a potential function. Therefore, we can define the potential function U as:

$$\begin{aligned} U(S,E,I,R,T) &= a_1S^2 + a_2E^2 + a_3I^2 + a_4R^2 + a_5T^2 \\ &\quad - \left[\Lambda S - \frac{\beta_1}{2}ES^2 - \frac{\beta_2}{2}IS^2 - \frac{\mu}{2}S^2E \right. \\ &\quad \left. - \left(\alpha E^2 - (\mu+\delta+\gamma+\gamma_1)I^2\right) - \mu R^2 - \mu T^2\right]. \end{aligned}$$

The potential function U captures the essential dynamics of the system and has a minimum value at the equilibrium point.

To prove Lyapunov stability, we need to show that $\left(\frac{dV}{dt} \le 0\right)$ almost surely (a.s.) for all (S,E,I,R,T) in the domain of interest, except at the equilibrium point. Taking expectations of dV and using the properties of Ito calculus, we obtain:

$$\begin{aligned} \mathbb{E}[dV] &= \mathbb{E}\left[2a_1S(\Lambda - (\beta_1E + \beta_2I)S - \mu S) + 2a_2E((\beta_1E + \beta_2I)S - (\alpha+\mu)E)\right. \\ &\quad \left. + 2a_3I(\alpha E - (\mu+\delta+\gamma+\gamma_1)I) + 2a_4R(\gamma I - \mu R) + 2a_5T(\gamma_1 I - \mu T)\right] \\ &\quad + \mathbb{E}\left[a_1\sigma S^2 + a_2\rho SE + a_3\theta SI + a_4\phi SR + a_5\psi ST\right]\mathbb{E}[dW]. \end{aligned}$$

Since the stochastic differentials dW have zero mean, the second term in the above equation becomes zero. Now, let us focus on the first term within the expectations. We can rewrite it as the derivative of the potential function U:

$$\begin{aligned} &\mathbb{E}\left[2a_1S(\Lambda - (\beta_1E + \beta_2I)S - \mu S) + 2a_2E((\beta_1E + \beta_2I)S - (\alpha+\mu)E)\right] \\ &+ \left[2a_3I(\alpha E - (\mu+\delta+\gamma+\gamma_1)I) + 2a_4R(\gamma I - \mu R) + 2a_5T(\gamma_1 I - \mu T)\right] \\ &= \frac{\partial U}{\partial S}\left[\Lambda - (\beta_1E + \beta_2I)S - \mu S\right] + \frac{\partial U}{\partial E}\left[(\beta_1E + \beta_2I)S - (\alpha+\mu)E\right] \\ &+ \frac{\partial U}{\partial I}\left[\alpha E - (\mu+\delta+\gamma+\gamma_1)I\right] + \frac{\partial U}{\partial R}\left[\gamma I - \mu R\right] + \frac{\partial U}{\partial T}\left[\gamma_1 I - \mu T\right]. \end{aligned}$$

We can simplify the above expression further by substituting the potential function U.

$$\begin{aligned} \mathbb{E}[dV] &= \frac{\partial U}{\partial S}\left[\Lambda - (\beta_1E + \beta_2I)S - \mu S\right] + \frac{\partial U}{\partial E}\left[(\beta_1E + \beta_2I)S - (\alpha+\mu)E\right] \\ &\quad + \frac{\partial U}{\partial I}\left[\alpha E - (\mu+\delta+\gamma+\gamma_1)I\right] + \frac{\partial U}{\partial R}\left[\gamma I - \mu R\right] + \frac{\partial U}{\partial T}\left[\gamma_1 I - \mu T\right] \\ &= \left(\frac{\partial U}{\partial S}\Lambda + \frac{\partial U}{\partial E}(\beta_1E + \beta_2I)S + \frac{\partial U}{\partial I}\alpha E + \frac{\partial U}{\partial R}\gamma I + \frac{\partial U}{\partial T}\gamma_1 I\right) \\ &\quad - \left(\frac{\partial U}{\partial S}(\beta_1E + \beta_2I)S + \frac{\partial U}{\partial E}(\alpha+\mu)E + \frac{\partial U}{\partial I}(\mu+\delta+\gamma+\gamma_1)I \right. \\ &\quad \left. + \frac{\partial U}{\partial R}\mu R + \frac{\partial U}{\partial T}\mu T\right) \\ &\quad - \mu\left(\frac{\partial U}{\partial S}S + \frac{\partial U}{\partial E}S + \frac{\partial U}{\partial I}I + \frac{\partial U}{\partial R}R + \frac{\partial U}{\partial T}T\right). \end{aligned}$$

To proceed, we need to compute the partial derivatives of the potential function U with respect to the variables S, E, I, R and T. Taking the partial derivatives, we have:

$$\begin{aligned} \frac{\partial U}{\partial S} &= 2a_1S - \left(\Lambda - \beta_1ES - \beta_2IS - \mu S\right), \\ \frac{\partial U}{\partial E} &= 2a_2E - \left(\beta_1ES - (\alpha+\mu)E\right), \\ \frac{\partial U}{\partial I} &= 2a_3I - \left(\beta_2IS + \alpha E - (\mu+\delta+\gamma+\gamma_1)I\right), \\ \frac{\partial U}{\partial R} &= 2a_4R - \gamma I, \\ \frac{\partial U}{\partial T} &= 2a_5T - \gamma_1 I. \end{aligned}$$

Now, substituting these partial derivatives into the expression for ($\mathbb{E}[dV]$), we obtain:

$$\begin{aligned} \mathbb{E}[dV] &= \left(\frac{\partial U}{\partial S}\Lambda + \frac{\partial U}{\partial E}(\beta_1E + \beta_2I)S + \frac{\partial U}{\partial I}\alpha E + \frac{\partial U}{\partial R}\gamma I + \frac{\partial U}{\partial T}\gamma_1 I\right) \\ &\quad - \left(\frac{\partial U}{\partial S}(\beta_1E + \beta_2I)S + \frac{\partial U}{\partial E}(\alpha+\mu)E + \frac{\partial U}{\partial I}(\mu+\delta+\gamma+\gamma_1)I \right. \\ &\quad \left. + \frac{\partial U}{\partial R}\mu R + \frac{\partial U}{\partial T}\mu T\right) \\ &\quad - \mu\left(\frac{\partial U}{\partial S}S + \frac{\partial U}{\partial E}S + \frac{\partial U}{\partial I}I + \frac{\partial U}{\partial R}R + \frac{\partial U}{\partial T}T\right) \\ &= -\left(\frac{\partial U}{\partial S}\beta_1E + \frac{\partial U}{\partial S}\beta_2I + \frac{\partial U}{\partial E}(\alpha+\mu)E + \frac{\partial U}{\partial I}(\mu+\delta+\gamma+\gamma_1)I \right. \\ &\quad \left. + \frac{\partial U}{\partial R}\mu R + \frac{\partial U}{\partial T}\mu T\right) \end{aligned}$$

which gives,

$$\begin{aligned} \mathbb{E}[dV] &= -\left(\frac{\partial U}{\partial S}\beta_1E + \frac{\partial U}{\partial S}\beta_2I + \frac{\partial U}{\partial E}(\alpha+\mu)E + \frac{\partial U}{\partial I}(\mu+\delta+\gamma+\gamma_1)I \right. \\ &\quad \left. + \frac{\partial U}{\partial R}\mu R + \frac{\partial U}{\partial T}\mu T\right). \end{aligned}$$

Now, substituting the partial derivatives of the potential function U that we computed earlier, we have:

$$\begin{aligned} \mathbb{E}[dV] &= -\left[\left(2a_1S - \left(\Lambda - \beta_1ES - \beta_2IS - \mu S\right)\right)\beta_1E \right. \\ &\quad + \left(2a_1S - \left(\Lambda - \beta_1ES - \beta_2IS - \mu S\right)\right)\beta_2I \\ &\quad + \left(2a_2E - \left(\beta_1ES - (\alpha+\mu)E\right)\right)(\alpha+\mu)E \\ &\quad + \left(2a_3I - \left(\beta_2IS + \alpha E - (\mu+\delta+\gamma+\gamma_1)I\right)\right)(\mu+\delta+\gamma+\gamma_1)I \\ &\quad \left. + 2a_4R\mu R + 2a_5T\mu T\right]. \end{aligned}$$

Simplifying further, we obtain:

$$\begin{aligned} \mathbb{E}[dV] &= -\left[2a_1\beta_1SE^2 + 2a_1\beta_2SI^2 + (\alpha+\mu)\beta_1SE^2 + (\alpha+\mu)\beta_2SI^2 \right. \\ &\quad \left. + 2a_2(\alpha+\mu)E^3 + 2a_3(\mu+\delta+\gamma+\gamma_1)I^3 + 2a_4\mu R^2 + 2a_5\mu T^2\right]. \end{aligned}$$

Now, to prove Lyapunov stability, we need to show that $\mathbb{E}[dV] \le 0$ for all $(S(t), E(t), I(t), R(t), T(t))$ in the domain of interest.

Since a_1, a_2, a_3, a_4, a_5 are positive constants, we can see that each term in the above expression is non-positive. Therefore, we have $\mathbb{E}[dV] \le 0$, indicating that the potential function V is decreasing on average.

Thus, by the Lyapunov stability theorem, the equilibrium point of the stochastic SEIRT model is Lyapunov stable. □

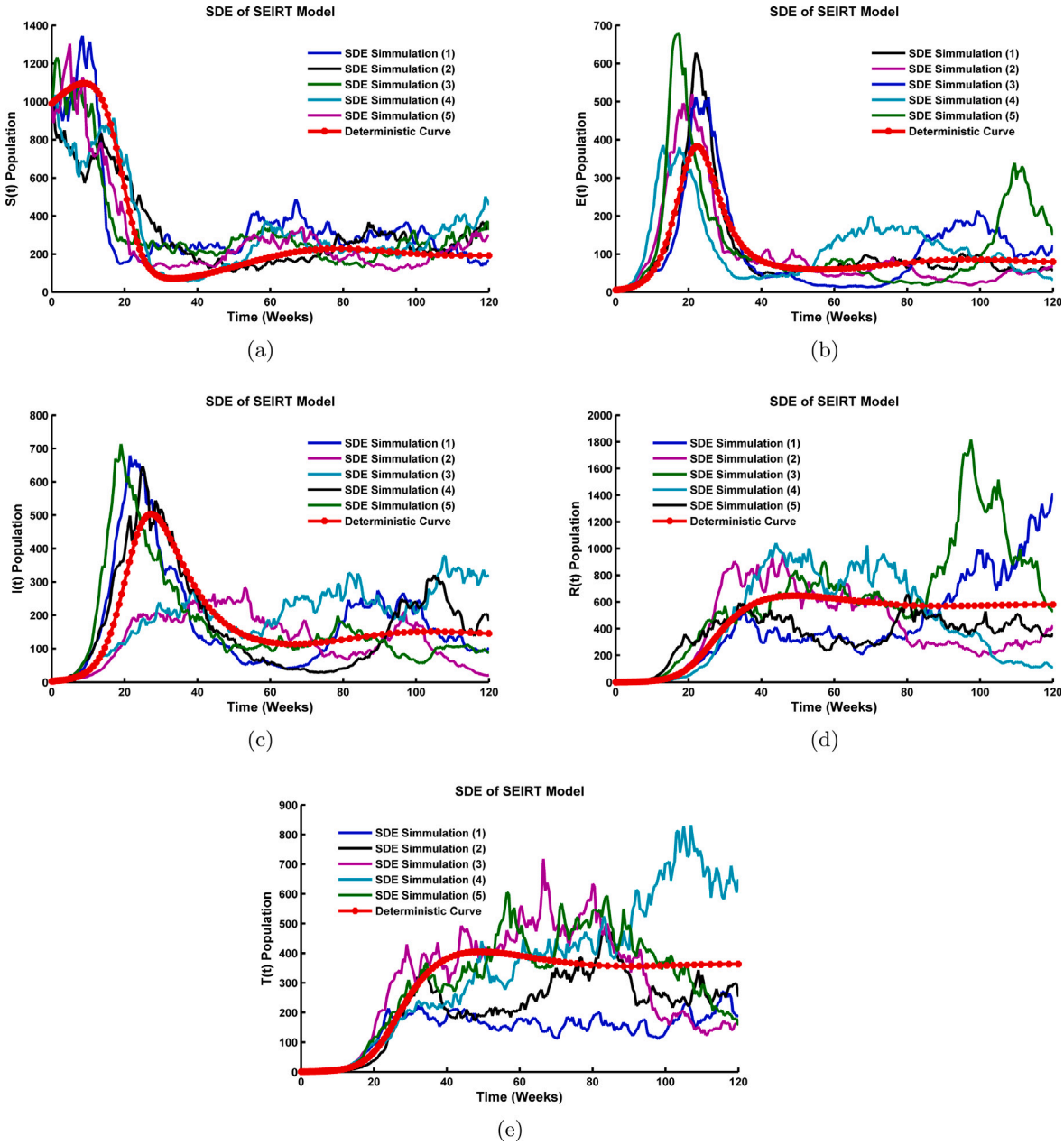


Fig. 7. The influenza SDE model's five sample routes are represented by the curves, where (a) behavior of susceptible $S(t)$, (b) behavior of exposed $E(t)$, (c) behavior of infected $I(t)$, (d) behavior of recovered $R(t)$, and (e) behavior of treated $T(t)$ population. The parameter values are $\beta_1 = 0.35$, $\beta_2 = 0.35$, $\gamma = 0.08$, $\gamma_1 = 0.05$, $\alpha = 0.25$, $\mu = 0.02$, $\delta = 0.01$, noise $= 0.1$..

7.1. SDE results and discussion

System (7.1) is the equivalent SDE model of (2.1). Within the schematics Fig. 7(a)–(e), We illustrate the framework (7.1) for susceptible, exposed, infected, recovered and treatment compartments. By choosing a reasonable amount of time for at least one event to occur, Fig. 7 displays five sample routes for each model compartment and a deterministic curve with a bold red line. For the graphs, in Fig. 7, we apply the CTMC model with the same parameter values and participant count as before. We have estimated the noise term $\sigma = 0.1$ for each compartments. The sample routes for exposed and infected humans indicate that, for the time being, the infection will subside and that the number of people who have recovered and received treatment will rise to a level at which our data collection is satisfactory.

In Figs. 7(a), (d) and (e) we have five sample paths for susceptible, recovered and treated individuals. With the time 0 to 20 weeks the susceptible population decreases gradually getting mixed with exposed and infected class. After 23 weeks, it slightly increases and fluctuates among 200 to 400. We also see that the recovered population gradually increase along - to 30 weeks. After that recovered population always remain in community at level 450 to 800. The treated individuals also increase gradually in time span 0 to 35 weeks. After, acquiring herd immunity and proper isolation to the exposed and infected compartments, the treatment class lies within 250 to 650.

With the progression of time, the number exposed and infected compartments increase gradually for 0 to 20 weeks. After that the affected individuals drop significantly for increasing recovery and treatment rates, also for gaining permanent immunity. More importantly, we noticed from the Figs. 5 and 7 that, notably, the SDE and CTMC curves are not differentiable, but they are continuous. Furthermore, the scenarios show that SDE offers a superior approximation and converges to the optimal solution more quickly than the CTMC model. Further,

the random fluctuation compared to the deterministic curves for each compartments can be visualized nicely by SDE formulation. Since a disease's recruitment rate plays a critical role in its persistence, it follows that disease propagation takes longer in bigger populations.

In this study, a detailed comparison of DTMC, CTMC, and SDE for modeling influenza disease dynamics is highlighted based on different strengths and trade-offs. DTMC models are effective when time is discretized into fixed intervals, making them easier to implement, but they can miss finer temporal changes, potentially reducing accuracy in continuous disease processes. CTMC, by contrast, allows for continuous-time transitions, providing a more accurate representation of the disease's real-time spread, though it can be more computationally demanding, especially for large populations. SDE models incorporate random fluctuations and stochastic variations in disease transmission, reflecting real-world uncertainties like super-spreader events or seasonal changes. While more accurate, SDE models are complex and require more advanced mathematical techniques. In influenza modeling, CTMC is ideal when time-continuous accuracy is essential, but SDE offers better performance when accounting for randomness in disease progression. DTMC may be useful for simpler, large-scale simulations with less emphasis on real-time detail.

8. Concluding remarks

We took into consideration the influenza model in our study, which uses the SEIRT epidemic model for host populations. For the influenza model, both deterministic and stochastic dynamics are created after accounting for all significant features of illness progression. We proved a number of theoretical results and illustrated them using the DTMC, CTMC, and SDE stochastic models. In essence, a satisfactory consensus is reached on the dynamic behavior of the stochastic and deterministic models. Additionally, it is shown that, after a predetermined amount of time, the number of exposed, sick persons will steadily decline. As a result, the deterministic and stochastic model we have in mind guarantees the efficacy of the plan to curtail and eventually stop disease outbreaks. These results are applicable to non-homogeneous stochastic epidemic models with periodic transmission. The season has a significant impact on a number of infectious diseases, such as leishmaniasis, influenza, chikungunya, dengue, Zika, malaria, and Lyme disease. The roles of host and pathogen reactions to seasonal cues in the environment are critical in defining the periodic transmission rates, which may then be used to determine the time-dependent risk of an outbreak by applying branching process approximation. Seasonal variations, the influence of climate change, and the prevalence of transmissible diseases are all important aspects of public health. We have used Lyapunov mapping to perform a global analysis in this work, which produced important analytical insights. distinct models, such as deterministic, continuous-time and discrete-time Markov Chains (CTMC and DTMC) and stochastic differential equations (SDEs), were shown to have distinct threshold quantities, $\mathcal{R}_0$. Our numerical results demonstrate that the interplay between two modeling techniques produces a convergence where stochastic methods and deterministic results closely match, providing a more realistic depiction of infection dynamics, transmission patterns, and intervention effects. Computational methods were used to validate theoretical conclusions, demonstrating that stochastic models offer a more thorough understanding of influenza epidemiology, preventative measures, and treatment outcomes for various situations. Additionally, the research indicates that long-term medicine and treatment can be maintained to control disease. The presented SEIRT model for influenza can be extended to other types of fractional derivatives by adjusting the order and type of the derivative, allowing flexibility in capturing different dynamic behaviors. This adaptability enhances the model's applicability across various epidemiological contexts. Future work will explore the dynamics of influenza and COVID-19 coinfection using the extended SEIRT model. The approaches and models discussed here, together with more complicated models, are helpful in developing experiments and testing ideas that will help to determine when disease-infected density control, preventative measures, and appropriate treatment policy are most significant.

CRediT authorship contribution statement

Kazi Mehedi Mohammad: Conceptualization, Data curation, Formal analysis, Methodology, Software, Validation, Writing – original draft. **Md. Kamrujjaman:** Conceptualization, Formal analysis, Funding acquisition, Investigation, Resources, Software, Supervision, Validation, Writing – original draft, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical approval

No consent is required to publish this manuscript.

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